

Algorithms and Combinatorics of Embedded graphs

Maths-CS track M2, UGE

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These are the lecture notes for the first half of the course on geometry, focusing on Algorithms and Combinatorics of Embedded Graphs.

This first half of the course focuses on embedded graphs: we start with *planar graphs* and then move on to *surface-embedded graphs*. These are graphs that can be drawn without crossings in the plane or in more complicated surfaces, see Figure 1. Therefore they form a *combinatorial* object with a *topological* constraint. The objective of the course is to *explore how topology interacts with combinatorics and algorithms* on this very natural class of objects. Some questions we will explore are:

- What are the combinatorial consequences of being planar? Are there combinatorial characterizations?
- Can one exploit planarity to design better algorithms for planar graphs than for general graphs?
- How do the previous answers generalize to graphs embedded in these more complicated surfaces?
- How to solve certain topological questions algorithmically on surfaces?

While the focus of the course will stay very theoretical (e.g., mostly with a theorem, lemma, proof structure), embedded graphs are of great interest for the practically-oriented mind, as they appear everywhere, for example in road networks (where underpasses and bridges can be modeled using additional topological features), chip design or the meshes that are ubiquitous in computer graphics or computer aided design. In all these applications, there is a strong need for a theoretical understanding of embedded graphs, as well as algorithmic primitives related to their topological features. Additionally, embedded graphs are an important lens to study graphs in general, since any graph can be embedded on some surfaces. This is especially the case in graph minor theory, where embedded graphs play an absolutely central role (but we will barely touch on this topic).

These lecture notes follow quite closely the material taught in class. This is actually their point, as in my experience lecture notes with too much content can easily get overwhelming (especially when one misses a class). Thus we will refrain from (too many) digressions and heavy references. The tone will be conversational. These lecture notes are mostly taken the notes of the lecture course given by Arnaud de Mesmay in 2023-2024, which are themselves heavily inspired by the lecture notes of a previous iteration of the class (by Éric Colin de Verdière), and those for a course on Computational Topology with Francis Lazarus. These notes is following the contents of the Lecture 2024-2025.

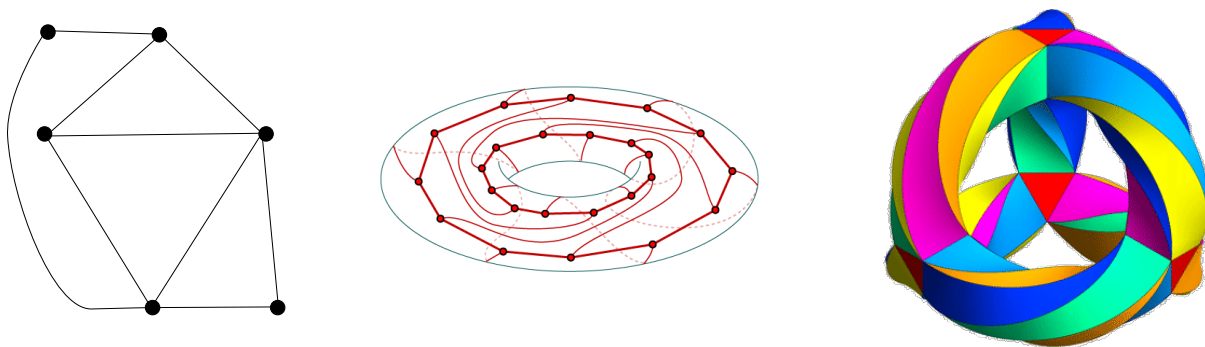


Figure 1: Graphs embedded in the plane, the torus, and the three-holed torus.

1 Planar Graphs

A graph is planar if it can be drawn on the plane so that no two edges intersect, except at their common endpoints. This very simple property has a very strong impact on the *combinatorics* of the graph: for example they are sparse (their number of edges is linear in the number of vertices), they can be colored with few colors; as well as on the *algorithms* that one can design for these graphs.

1.1 Point-set topology, graphs and embeddings

We will use basic notions of point set topology. For the unfamiliar reader, this is like metric topology, but without a metric (!). This is very relevant for because in most cases, we will not care about the Euclidean metric on the plane (or any other), and a vague notion of neighborhood will be enough. A *topological space* is a set X with a collection of subsets of X , called *open sets*, such that:

- the empty set and X are open.
- any union of open sets is open.
- Any finite intersection of open sets is open.

Any metric space gives rise to a topological space by taking as open sets the smallest family of sets containing the open (in the metric sense) balls and satisfying the above axioms. A map f between two topological spaces is *continuous* if the inverse image of any open set by f is open. A map f is a *homeomorphism* if it is continuous, bijective, and if its inverse f^{-1} is also continuous. Homeomorphic spaces throughout these notes will generally be considered to be *the same*, and thus identified without precaution. Therefore, when we talk about *the sphere*, we talk about any space homeomorphic to the 2-sphere $\mathbb{S}^2 = \{x, y, z \mid x^2 + y^2 + z^2 = 1\}$ with its standard topology.

A *graph* $G = (V, E)$ is defined by a set $V = V(G)$ of vertices and a set $E = E(G)$ of edges where each edge is associated one or two vertices, called its *endpoints*. A *loop* is an edge with a single endpoint. Edges sharing the same endpoints are said *parallel* and define a *multiple edge*. A graph without loops or multiple edges is called *simple*. In a simple graph every edge is identified unambiguously with the pair of its endpoints. A *path* is an alternating sequence of vertices and arcs such that every arc is preceded by its origin vertex and followed by the origin of its opposite arc. A path may have repeated vertices (beware that this is not standard, and usually called a *walk* in graph theory books). Two or more paths are *independent* if none

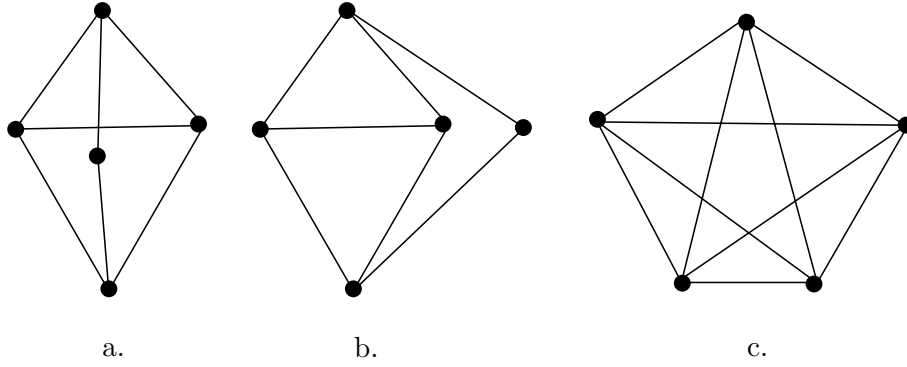


Figure 2: a. A planar graph (!). b. The same planar graph, this time embedded in the plane, i.e., a plane graph. c. A non-planar graph

contains an inner vertex of another. A **circuit** is a closed path, i.e., a path whose first and last vertex coincide. A **cycle** is a simple circuit (without repeated vertices). We will restrict to finite graphs for which V and E are finite sets.

The Euclidean distance in the plane $\mathbb{R}^2$ induces the usual topology where a subset $X \subset \mathbb{R}^2$ is open if every of its points is contained in a ball that is itself included in X . The **closure** $\bar{X}$ of X is the set of limit points of sequences of points of X . The **interior** $\overset{\circ}{X}$ of X is the union of the open balls contained in X . The boundary ∂X is the closure minus the interior. An **embedding** of a non-loop edge in the plane is a topological embedding (i.e., an injective map) of the segment $[0, 1]$ into $\mathbb{R}^2$. Likewise, an embedding of a loop-edge is an embedding of the circle $S^1 = \mathbb{R}/\mathbb{Z}$. Such an embedding of the circle is called a **simple closed curve** in the plane. An **embedding** of a finite graph $G = (V, E)$ in the plane is defined by a bijective map $V \rightarrow \mathbb{R}^2$ and, for each edge $e \in E$, by an embedding of e sending $\{0, 1\}$ to e 's endpoints such that the relative interior of e (the image of $]0, 1[$) is disjoint from other edge embeddings and vertices¹. Note that this definition can be (and will be) generalized to other topological spaces than $\mathbb{R}^2$.

A graph is **planar** if it has an embedding into the plane, see Figure 2. The data of the planar graph and a specific embedding is called a **plane** graph, although it is customary to mix up everything and also call it a planar graph. We consider two plane graphs to be the same if they are homeomorphic, i.e., if there exists a homeomorphism of the plane sending one to the other.

Exercise 1.1. 1. Any graph can be embedded in $\mathbb{R}^3$.

2. A graph can be embedded in the plane if and only if it can be embedded in the sphere. (Hint: prove that the plane is homeomorphic to the sphere with one arbitrary point removed).

The second exercise allows us to jump at will between embeddings in the plane and on the sphere, which we will constantly do.

1.2 The Jordan curve theorem

The main additional structure that a planar graph has, compared to an abstract one, is that it has faces. A **face** of a graph embedding is a connected component of the complement of the image (via the embedding) of the vertices and edges of the graph. Then the basic properties of

¹In other words, this is a topological embedding of the quotient space $(V \sqcup [0, 1] \times E) / \sim$, where $\sim$ identifies edge extremities $(0, e)$ and $(1, e)$ with the corresponding vertices.

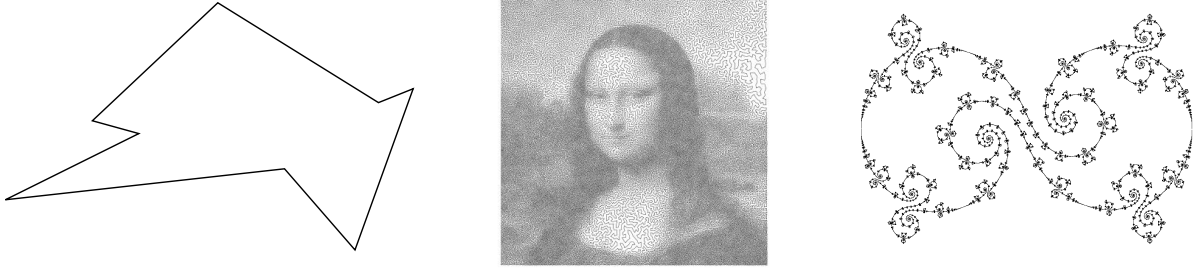


Figure 3: Three simple closed curves in the plane

planar graphs follow from the fact that a simple closed curve in the plane is the boundary of exactly two components, one of which is a disk:

Theorem 1.2 (Jordan curve theorem). *Let C be a simple closed curve in the plane. Then its complement has two connected components, one of which is bounded, and each of which has C as a boundary.*

This obvious fact is famously painful to prove. One reason is that simple closed curves can be very pathological, like fractals, see for example Figure 3. In the simpler case that C is polygonal, i.e., a concatenation of segments, the proof is much simpler.

Proof in the polygonal case. Since C is contained in a compact ball, its complement has exactly one unbounded component. Define the *horizontal rightward* direction $\vec{h}$ as some fixed direction transverse to the all the line segments of C . Rotate the plane so that the direction that we have chosen is actually horizontal. For every segment s of C we let $\underline{s}$ be the lower half-open segment obtained from s by removing its upper endpoint. We also denote by h_p the ray with direction $\vec{h}$ starting at a point $p \in \mathbb{R}^2$. We consider the *parity function* $\pi : \mathbb{R}^2 \setminus C \rightarrow \{ \text{even, odd} \}$ that counts the parity of the number of lower half-open segments of C intersected by a ray:

$$\pi(p) := \text{parity of } |\{ \text{segment } s \text{ of } C \mid h_p \cap \underline{s} \neq \emptyset \}|$$

Every $p \in \mathbb{R}^2 \setminus C$ is the center of small disk D_p over which π is constant. Indeed, let S_p be the set of segments (of C) that avoids h_p , let S'_p be the set of segments whose interior crosses h_p and let S''_p be the set of segments whose lower endpoint lies on h_p . If D_p is sufficiently small, then for every $q \in D_p$ we have $S_q = S_p$, $S'_q = S'_p$ and the parity of $|S''_q|$ and $|S''_p|$ is the same, see Figure 4. It follows that $\pi(q) = \pi(p)$.

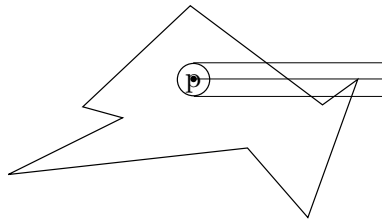


Figure 4: Polygonal Jordan curve theorem

Since π is locally constant, it must be constant over each connected component of $\mathbb{R}^2 \setminus C$. Moreover, the parity function must take distinct values on points close to C that lie on a same horizontal but on each side of C . It follows that $\mathbb{R}^2 \setminus C$ has at least two components. To see that $\mathbb{R}^2 \setminus C$ has at most two components consider a small disk D centered at a point interior to a segment s of C . Then $D \setminus C = D \setminus s$ has two components. Moreover, any point in $\mathbb{R}^2 \setminus C$

can be joined to one of these components by a polygonal path that avoids C : first come close to C with a straight line and then follow C in parallel until D is reached. Finally, it is easily seen by similar arguments as above that every point of C is in the closure of both components of $\mathbb{R}^2 \setminus C$. $\square$

A stronger version of this theorem, which we will not prove either, shows that one of the two components is homeomorphic to an open disk.

Theorem 1.3 (Jordan-Schoenflies Theorem). *Let C be a simple closed curve in the plane. Then its complement has two connected components, one of which is a topological disk, and each of which has C as a boundary.*

One (not that immediate) consequence of this theorem is that in an embedding of a connected graph, all of the faces except one are open disks (or equivalently, on a sphere, all of the faces are open disks). Such a plane graph is called a **cellular embedding**. This definition is not very interesting for now but will get more important when embedding graphs on other surfaces. Each face can be described by a **facial walk**, which is the circuit on the graph describing the boundary of the face. Note that a facial walk may have repeated vertices and edges (see Figure 5, left), but this does not happen when the graph is connected enough. A graph G is **2-connected** if for any vertex x , $G \setminus x$ is connected.

Lemma 1.4. *In a 2-connected planar graph, facial walks are cycles of the graphs (i.e., without repeated vertices and edges).*

Proof. Assume otherwise, let F be an open disk bounded by a facial walk that uses the same vertex x twice. Then $F \cup \{x\}$ contains a Jordan curve separating the graph, contradicting the 2-connectedness assumption. $\square$

The **dual graph** of a cellular graph embedding $G = (V, E)$ on $\mathbb{S}^2$ is a plane graph G^* defined as follows: put one vertex f^* of G^* in the interior of each face of G , and for each edge e of G , create a dual edge in G^* crossing e and no other edge of G . The dual graph is also cellular, and is, up to homeomorphism, uniquely defined by the above. Note that the dual graph crucially depends on the embedding, and that duality does not preserve simplicity. As the name suggests, duality is an involution: $G^{**} = G$. See Figure 5, right).

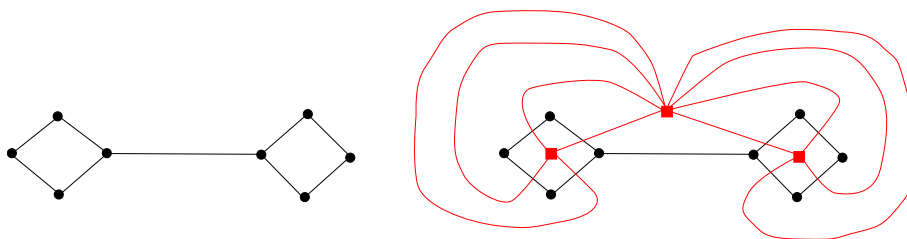


Figure 5: A cellularly embedded graph and its dual

This allows us to think of a plane graph as a collection of open disks glued together with a combinatorial data, which is convenient both for combinatorial and algorithmic purposes.

1.3 Euler's formula

Euler's formula shows that the number of vertices, edges and faces of a graph are related by a linear equation. It has a rich history, of which we will only say that it was likely not first discovered by Euler.

Lemma 1.5. *Let G be a plane graph. If v is a vertex of degree one in G then $G - v$ and G have the same number of faces.*

Proof. We denote by e the edge incident to v in G . Every face of G is contained in a face of $G - v$. Moreover, the relative interior of (the embedding of) e is contained in a face f of $G - v$. Hence, every other face of $G - v$ is also a face of G . It remains to count the number of faces of G in f . Let p, p' be two points in $f \setminus e$. There is a path in f connecting p and p' . This path may intersect e , but we may avoid this intersection by considering a detour in a small neighborhood N_e of e in f (indeed, $N_e \setminus e$ is connected). It follows that p and p' belong to a same component of $f \setminus e$. We conclude that G has only one face in f , so that G and $G - v$ have the same number of faces. $\square$

Theorem 1.6 (Euler's formula). *Let $|V|, |E|$ and $|F|$ be the number of vertices, edges and faces of a connected plane graph G . Then,*

$$|V| - |E| + |F| = 2$$

Proof. We argue by induction on $|E|$. If G has no edges then it has a single vertex and the above formula is trivial. Otherwise, suppose that G has a vertex v of degree one. Then by Lemma 1.5, G has the same number of faces as $G - v$. Note that G has one vertex more and one edge more than $G - v$. By the induction hypothesis we can apply Euler's formula to $G - v$, from which we immediately infer the validity of Euler's formula for G . If every vertex of G has degree at least two, then G contains a cycle C . Let e be an edge of C . We claim that G has one face more than $G - e$. This will allow to conclude the theorem by applying Euler's formula to $G - e$, noting that G has the same number of vertices but one edge less than $G - e$. By the Jordan curve theorem 1.2, C cuts the plane into two components bounded by C . Since $G = C \cup (G - e)$, every face of G is included in the intersection of a face of C and a face of $G - e$. Let f be the face of $G - e$ containing the relative interior of e . Every other face of $G - e$ does not meet C , hence is also a face of G . Since f intersect the two faces of C (both bounded by e), G has at least one face more than $G - e$. By considering a small tubular neighborhood of e in f , one shows by an already seen argument that $f \setminus e$ has at most two components. It follows that f contains exactly two faces of G , which concludes the claim. $\square$

To illustrate the power of this formula, here is a standard application.

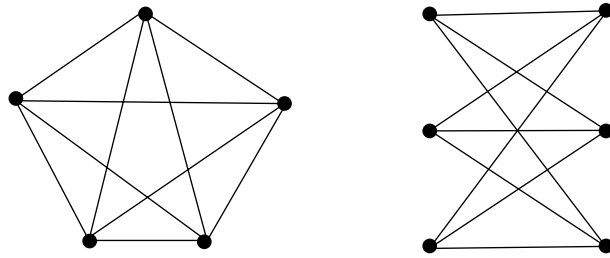


Figure 6: The graphs K_5 and $K_{3,3}$.

Theorem 1.7. *The graphs K_5 and $K_{3,3}$ (see Figure 6) are not planar.*

Proof. In any planar embedding of a graph, each edge is adjacent to two faces (perhaps twice the same). Since K_5 is simple, each face is adjacent to at least 3 edges. In any planar embedding of K_5 , the Euler formula would give $|V| - |E| + |F| = 2$, and by the previous two observations, we have $3|F| \leq 2|E|$, therefore $|V| - |E|/3 \geq 2$. But plugging in the values gives $5/3 \geq 2$, a contradiction. For $K_{3,3}$, we observe that any facial walk in a planar embedding would have at

least 4 edges because the graph is bipartite. This leads to $|V| - |E|/2 \geq 2$, e.g., $3/2 \geq 2$, another contradiction. $\square$

The following two applications, very close to the proof we have just seen, are absolutely key.

Exercise 1.8. Every simple planar graph G with $n \geq 3$ vertices has at most $3n - 6$ edges.

Exercise 1.9. Every simple planar graph has a vertex with degree less than 6.

The first one shows that simple planar graphs are sparse, i.e., verify $|E| = O(|V|)$. Therefore, when talking about the complexity of a simple connected planar graph, we can refer either to its vertices or edges.

The second exercise has the following nice application. A k -coloring of a graph is a map $c : V \rightarrow \{1, \dots, k\}$ so that two adjacent vertices are mapped to different colors.

Proposition 1.10. *Planar graphs are 6-colorable.*

Proof. We prove the result by induction on the number of vertices. For low values, this is immediate. For the induction step, pick a vertex z of degree less than 6, and color inductively $G \setminus z$. Then the five neighbors have at most 5 different colors, and we can color z with one of the remaining colors. $\square$

Exercise 1.11. Planar graphs are 5-colorable. Hint: look at paths connecting non-adjacent neighbors of a degree 5 vertex.

As is well known, planar graphs are actually 4-colorable, and we will not prove this in the course.

1.4 Planar graph and k-connectivity

Planar graphs define faces as connected components of $\mathbb{R}^2 \setminus G$. Using Euler's formula and Jordan curve theorem, we have a well understanding on the geometry of the faces in function of their connectivity. Here a list of theorem proved into the lecture.

Exercise 1.12. Let G be a 4-vertex graph which contains a K_3 subgraph. Prove that for any piecewise linear drawing in $\mathbb{R}^2$, the image of K_3 is the boundary of one face.

Exercise 1.13. Find a 5-vertex graph which contains a K_3 such that there is no embedding of G where K_3 bound a face.

Definition 1.14. *An outerplanar graph is a planar graph such that there exists an embedding of G where all vertices are incident to the exterior face.*

Exercise 1.15. Find a such graph with 4 faces.

Proposition 1.16. K_4 and $K_{2,3}$ are not outerplanar.

Theorem 1.17. *The graph G is outerplanar if and only if G does not contains a subdivision of K_4 or $K_{2,3}$ as a subgraph*

Application: Maximal planar graph.

Definition 1.18. G is a maximal planar graph if for all additional edge e where end points are in V and $e \in E$, the $G + e$ is not planar.

Definition 1.19. A triangulation is an embedding of a planar graph with any face of G is bounded by a K_3 cycle.

Theorem 1.20. *Every embedding of a planar maximal graph is a triangulation.*

Lemma 1.21. *Let G be an embedded planar graph. If G is 2-connected then every face of G is bounded by a cycle.*

Lemma 1.22. *If G is 2-connected, then every edge is in a cycle of the graph.*

Exercise 1.23. Prove that if a planar graph has all its faces bounded by a cycle then G is 2-connected.

Proposition 1.24. *Every maximal planar graph with at least 4-vertex is 3-connected.*

Definition 1.25. *A subgraph $H \subset G$ is non separating if $G \setminus H$ is connected*

Definition 1.26. *An induced cycle is a cycle without chord edge.*

Theorem 1.27. *Let G be a 3-connected planar graph. A cycle $C \subset G$ bounds a face if and only if C is a non separating induced cycle.*

Theorem 1.28. (Menger) *Let G be a 3-connected planar graph and two vertices u and v . Then there exists at least three independent paths connecting u and v which are not intersecting beside the end points u and v .*

Exercise 1.29. 1. A graph embedded on the plane is a triangulation if every face (including the outerface) is adjacent to exactly three edges. Show that if G is a triangulation, its numbers of vertices and faces satisfy $2|V| = 4 + |F|$.

2. Let G be a simple planar graph, and suppose that we arbitrarily color each edge of G either blue or red. Prove that for any planar embedding of G , there exists a vertex around which the incident red edges are consecutive. *Hint: Add edges to G to obtain a triangulation G' , then define a well-chosen bipartite graph out of the vertices and faces of G' , look at the degree of its vertices and apply the previous question.*

Solution of the exercise:

1. In a triangulation, every face is adjacent to exactly three edges, so by double-counting edges, we obtain that $3|F| = 2|E|$. We plug that into the Euler formula $|V| - |E| + |F| = 2$ to obtain that $|V| - |F|/2 = 2$ and thus $2|V| = 4 + |F|$.
2. First, we add edges to G until we obtain a triangulation G' and color its edges arbitrarily. If we prove the property for G' , it also proves it for G : indeed, if a vertex in G' has all its incident red edges consecutive, this is also the case for the same vertex in G , since it just has less edges. Now, let us define a graph H as follows. It has two types of vertices, either vertices of G' (vertices of the *first* type) or faces of G' (vertices of the *second* type). There is an edge between a first-type vertex and a second-type vertex if the corresponding vertex and face are incident in G' , and if the two edges of G' that are adjacent to that incidence have different colors. Then we remove the vertices of the second type with no adjacent edges. This graph H has the following properties: (i) All the vertices of the second type have degree two and (ii) If G has no vertex with all incident red edges consecutive, then the vertices of the first type all have degree at least four (since they see at least at least alternation red-blue-red-blue). But the number of vertices of the first type in H is $|V(G')|$ and the number of vertices of the second type in H is at most $|F(G')|$ (recall that we might have thrown some away). Therefore, by double counting the edges in this bipartite graph, we have that $4|V(G')| \geq 2|V(H)|$. But by the previous exercise, $2|V(H)| = 4|F(H)| + 4$. This is a contradiction, which concludes the exercise.

1.5 Kuratowski's Theorem

We say that H is a **subdivision** of G if H is obtained by replacing the edges of G by independent simple paths of one or more edges. Obviously, a subdivision of a non-planar graph is also non-planar. It follows from Theorem 1.7 that a planar graph cannot have a subdivision of K_5 or $K_{3,3}$ as a subgraph. It turns out that the converse is also true.

Theorem 1.30 (Kuratowski, 1929). *A graph is planar if and only if it does not contain a subdivision of K_5 or $K_{3,3}$ as a subgraph.*

This is a striking instance of the interplay between the topology and the combinatorics: our topological constraint can be replaced by a purely combinatorial one, and vice versa. In the following K_5 and $K_{3,3}$ are called forbidden graphs.

Contracting an edge e in a graph G means removing it and identifying its two endpoints. When the graph is embedded, the contraction can be done while preserving the embedding. We denote this operation by G/e . A 3-connected graph is a graph that stays connected when one removes an arbitrary pair of vertices.

Exercise 1.31. Let G be a graph such that G/e contains a subdivision of K_5 or $K_{3,3}$. Then G also contains a such a subdivision. (Hint: this might not be the same forbidden graph!).

The proof of Kuratowski's theorem is very nice and clean in the 3-connected case, and requires somewhat tedious machinery to reduce to that case. This machinery is summarized in the following two lemmas, whose proof is skipped in class.

Lemma 1.32. *Any 3-connected graph G with at least five vertices contains an edge e such that G/e is 3-connected.*

Lemma 1.33. *Let G be a graph with at least five vertices, containing no subdivision of K_5 nor of $K_{3,3}$ and such that the addition of any edge between non-adjacent vertices creates such a subdivision. Then G is 3-connected.*

Proposition 1.34 (Kuratowski's theorem for 3-connected graphs). *A 3-connected graph G without any subgraph isomorphic to a subdivision of a forbidden graph is a planar graph.*

Proof. We prove the contrapositive: any 3-connected non-planar graph contains a subdivision of K_5 or $K_{3,3}$ as a subgraph. In order to do so, let us pick G a contraction-minimal non-planar graph, i.e., a graph such that for any edge, contracting the edge e yields a planar graph. We prove that G contains a subdivision of K_5 or $K_{3,3}$ as a subgraph, which proves the theorem, since any non-planar graph contains a contraction-minimal non-planar graph as a subgraph, and the Exercise 1.31 guarantees that this property will still be satisfied after uncontracting.

Let us pick an edge $e = xy$ of G so that G/e is 3-connected, as guaranteed by Lemma 1.32. Then G/e is planar. In the graph G/e , e corresponds to a single vertex z , and $G/e \setminus z$ is 2-connected. Hence the face that contains z is a cycle C of the graph, by Lemma 1.4. Now we try to uncontract z while preserving planarity: this will fail either if x and y have at least three adjacent vertices in common, see Figure 7, right, or if x and y are adjacent to four vertices that *interleave*, that is, if there exist vertices x_1, x_2 adjacent to x , and y_1 and y_2 adjacent to y , that are cyclically ordered as x_1, y_1, x_2, y_2 on C , see Figure 7, left. In the first case, G contains a subdivision of K_5 , and in the second case it contains a subdivision of $K_{3,3}$. $\square$

Proof of Kuratowski's theorem. One direction has already been proved: a graph that contains a subdivision of K_5 or $K_{3,3}$ cannot be planar. For the reverse direction, let G be a graph that

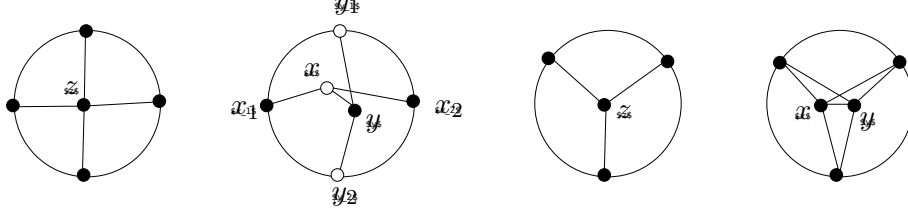


Figure 7: The two cases to find subdivisions of forbidden graphs.

does not contain a subdivision of K_5 or $K_{3,3}$. Add edges to G until it is maximal with this property, i.e., adding any new edge yields a subdivision of K_5 or $K_{3,3}$. By Lemma 1.33, the new graph G' is 3-connected, and thus by Proposition 1.34, it is planar. Since G is obtained from G' by removing edges, it is also planar, concluding the proof. $\square$

Now for the missing proofs, in a quite hasty style:

Proof of Lemma 1.32. Let $e = xy$ be an edge of G . If G/e is not 3-connected, then G contains a vertex z such that $G \setminus \{x, y, z\}$ is disconnected. Let G_1 be the smallest component of $G \setminus \{x, y, z\}$. Then z is joined to G_1 by an edge $e_1 = zu$. If G/e_1 is not 3-connected, then there is a vertex v such that $G \setminus \{z, u, v\}$ is disconnected, and since $G \setminus z$ is 2-connected, v is in $G_1 \cup \{x, y\}$ and the smallest component is a subgraph of G_1 . So we continue until we get the edge that we are looking for. $\square$

Proof of Lemma 1.33. We pick a minimal counter-example: a smallest graph G with at least 4 vertices that is not 3-connected, containing no subdivision of a forbidden graph and such that the addition of any edge between non-adjacent vertices creates such a subdivision.

We claim that G is 2-connected. Otherwise, there is a cut vertex x , adjacent to vertices y_1 and y_2 , each on one side. Adding an edge between y_1 and y_2 creates a subdivision K of a forbidden graph. Since K_5 and $K_{3,3}$ are both 3-connected, all the vertices of degree at least 3 of K are on the same side. Then the only way for K to use the other side is if a path uses the edge y_1y_2 and comes back via x , but it might as well use the edge y_ix , and so K was already present in G .

Now, if G is not 3-connected, there are two vertices x and y disconnecting it. We claim that the edge xy must be in G . Otherwise, adding it creates a subdivision K of a forbidden graph, and by the same previous argument, all of its vertices of degree at least 3 must be on the same side. Then we can reroute a path going through the other side with the edge xy , and so K was already present in G .

So now we are in the case where $\{x, y\}$ disconnects G into G_1 and G_2 and the edge xy is in G . Adding some edge in G_i creates a subdivision of a forbidden graph, which, by the same arguments as before, is contained in G_i . By minimality of G , G_i is 3-connected or has at most three vertices. By Proposition 1.34, they are both planar, pick one of their embeddings. Let $z_i \neq x, y$ be a vertex of a face F_i of G_i bounded by xy . Note that F_i must be a triangle, as otherwise one could add a diagonal within the face, preserving planarity and thus also the fact that it contains no subdivision of a forbidden graph. Then adding z_1z_2 creates a subdivision of a forbidden graph K .

At this stage, in the graph $G + z_1z_2$, there are now three vertices connecting G_1 and G_2 , so it might be that the vertices of degree at least 3 of K are not all contained in the same side. Actually, this has to be the case, otherwise, some path uses z_1z_2 and either x and y , and thus can be rerouted to z_ix or z_iy . So K has at least one vertex on each side. If K is a subdivision of K_5 , then there are four independent paths connecting those two vertices, which

is not possible here. So K is a subdivision of $K_{3,3}$, and by a similar argument, five of its vertices are on one side, say G_1 and one on the other side. But then we could as well put that last vertex at some place p inside F_1 and connect it to x , y and z_1 . This graph $G_1 + \{px, py, pz\}$ is planar by construction, but contains a subdivision of $K_{3,3}$. This last contradiction concludes the proof. $\square$

A *minor* of a graph G is any graph obtained from a subgraph of G by contracting a subset of its edges. If G contains a subdivision of H then H is a minor of G , but the converse is not true (can you find a counter-example?). Nevertheless, a similar theorem to Kuratowski's hold for minors:

Theorem 1.35 (Wagner, 1937). *A graph G is planar if and only if none of K_5 and $K_{3,3}$ is a minor of G .*

Proof. If G is not planar, then by Kuratowski's theorem it contains a subdivision of K_5 or $K_{3,3}$ as a subgraph, and thus a K_5 or a $K_{3,3}$ minor. For the converse direction, a minor of a planar graph is planar, so this follows from Theorem 1.7. $\square$

Wagner's theorem admits generalizations to graphs embedded on other surfaces, and in what constitutes a jewel of structural graph theory, to any minor-closed family of graphs.

The following theorem is a strong application of Kuratowski's theorem. Two edges are *independent* if they do not share a vertex.

Theorem 1.36 (Strong Hanani-Tutte theorem, 1934). *Any drawing of a non-planar graph contains two independent edges that cross an odd number of times.*

Conversely, if one can draw a graph so that all independent edges cross an even number of times, then the graph is planar.

Proof. We first claim that it suffices to prove the theorem for K_5 and $K_{3,3}$. Indeed, let G be a non-planar graph. By Kuratowski's theorem, it contains a subdivision of K_5 or $K_{3,3}$. If the theorem is proved for those, then we can find a pair of disjoint paths in our graph G that cross an odd number of times. This means that there exists a pair of independent edges that cross an odd number of times.

In order to prove the theorem for K_5 and $K_{3,3}$, we prove the stronger result that in any drawing of these two graphs, the sum of the number of crossings over all independent pairs of edges is odd. The reason is that this quantity mod 2 does not actually depend on the drawing. Indeed, given two drawings of K_5 or $K_{3,3}$, one can continuously morph one drawing into the other by moving the vertices so that they match, and then matching the edges. This can be done in a way that all the crossings stay transverse except at a finite number of singularities: when two edges start crossing, when the pattern of intersection between edges changes or when an edge passes on the other side of a vertex². Note that whether a pair of edges crosses an odd number of times only changes when an edge passes to the other side of a vertex. But in the case of K_5 and $K_{3,3}$, the evenness of the sum of these crossings actually does not change: for K_5 it changes by 4, and for $K_{3,3}$ it changes by 2. We conclude the proof by exhibiting drawings of K_5 and $K_{3,3}$ where this sum is odd, and we have already done so in Figure 6. $\square$

²It is not that obvious how to do so for continuous edges which might be pathological, but up to homeomorphism one can assume that they are polygonal, or even linear thanks to Theorem 1.37

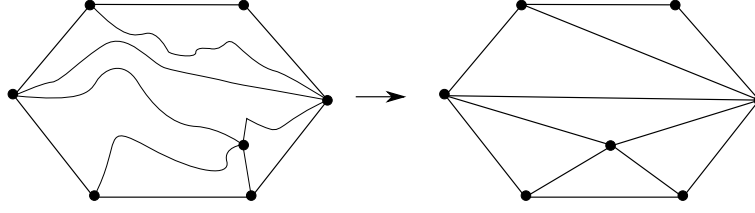


Figure 8: The Tutte magic

1.6 The Tutte barycentric embedding

In this section, we explain a very clean conceptual way to find a nice embedding of a planar graph, with just the knowledge of a facial cycle of the graph. We will assume that the graph is simple and *triangulated*, that is, that each facial cycle, except possibly the unbounded one³, is adjacent to three edges. Note that if one is given a planar embedding of a graph, one can add edges inside the faces of degree more than 3, subdividing those until the graph is triangulated. By the second exercise in the homework, a triangulated graph with at least 4 vertices is 3-connected.

Then the Tutte embedding for a triangulated⁴ graph proceeds as follows

1. Pick a facial cycle of G , which will bound the outer face.
2. Nail all its vertices in some strictly convex position on a wall.
3. Let the other vertices move freely, and put springs in the place of the edges, each with some arbitrary positive stiffness.
4. Let the system relax.
5. Enjoy the nice planar embedding of the graph at equilibrium.

This is illustrated in Figure 8.

More formally, denoting the vertices not on the outer face by V_I , we have

Theorem 1.37 (Tutte, 1963). *Let G be a triangulated graph and C be the facial cycle bounding the outer face. Every strictly convex embedding of the vertices of C extends to a unique map $\tau : V \rightarrow \mathbb{R}^2$ such that for every internal vertex v , its image $\tau(v)$ is the convex combination of the image of its neighbors $N(v)$ with weights λ_{vw} , for $w \in N(v)$:*

$$\forall v \in V_I, \quad \sum_{w \in N(v)} \lambda_{vw} (\tau(v) - \tau(w)) = 0. \quad (1.1)$$

Moreover, τ induces an embedding of G by connecting the images of every pair of neighbor vertices with line segments.

When one takes symmetric weights $\lambda_{vw} = \lambda_{wv}$, the weights can be interpreted as the stiffness of springs and the system of equations of this theorem is the equilibrium condition of the physical system (obtained for example by differentiating the energy). Even more simply, taking all the weights equal to one puts each vertex that is not on the outer face at the barycenter of its neighbors. The theorem then stipulates that this very intuitive placement yields a planar embedding. In particular, one obtains:

³There is no consensus on this definition, sometimes the outer one is also forced to be triangular, sometimes not

⁴It actually suffices for the graph to be 3-connected for this to work, we do the triangulated case for simplicity.

Theorem 1.38 (Fàry-Stein-Wagner). *Every planar graph can be drawn in the plane with straight line edges.*

There are easier proofs of this theorem, but I like this one because of how conceptual it is. The following notes will be alas pretty devoid of pictures because it is very hard to picture things that do not happen.

Throughout the proof, we number the internal vertices from 1 to k and the remaining ones from $k + 1$ to n , and to homogenize notation, we consider the weight λ_{ij} for two non-adjacent vertices to be zero (this corresponds to a spring that does not pull). We first look at the system of equations 1.1:

Lemma 1.39. *The system of linear equations 1.1 has a unique solution.*

Proof. The unknowns are the positions of the vertices $\tau(i)$ for i from 1 to k .

(1.1) can be written

$$\Lambda \begin{bmatrix} \tau_1 \\ \vdots \\ \tau_k \end{bmatrix} = \begin{bmatrix} \sum_{j>k} \lambda_{1j} \tau_j \\ \vdots \\ \sum_{j>k} \lambda_{kj} \tau_j \end{bmatrix}$$

where τ_i stands for $\tau(i)$ and

$$\Lambda = \begin{bmatrix} \sum_{j=1}^n \lambda_{1j} & -\lambda_{12} & \dots & -\lambda_{1k} \\ -\lambda_{21} & \sum_{j=1}^n \lambda_{2j} & \dots & -\lambda_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ -\lambda_{k1} & -\lambda_{k2} & \dots & \sum_{j=1}^n \lambda_{kj} \end{bmatrix}$$

Beware that the notations used are a bit misleading, as each τ_i is two-dimensional. To conclude, it suffices to prove that Λ is invertible. It follows from the fact that it is close to being diagonally dominant, i.e., each diagonal entry is bigger than the sum of the other entries on the same row. A proof goes as follows: if there exists x such that $\Lambda x = 0$, take x_i to be one of its components of maximal absolute value. Then the only way for $(\Lambda x)_i = \sum_{j \in N(i)} \lambda_{ij}(x_i - x_j)$ to be zero is that for all j such that $\lambda_{ij} > 0$, we have $x_j = x_i$. By connectivity of G , this extends to all the x_j , in particular those for vertices adjacent to a boundary vertex. But for those, the row is diagonally dominant, and thus $x_j = 0$. We conclude that Λ is invertible. $\square$

Note that this theorem is readily algorithmic, as solving this system directly gives the coordinates of the vertices.

For an interior vertex v placed at $\tau(v)$, let $h : \mathbb{R}^2 \rightarrow \mathbb{R}$ denote an affine function vanishing on $h(\tau(v))$. We say that v is ***h-inactive*** if all its neighbors have the same h -value. Otherwise, it is ***h-active***. If a vertex is active, since it is a barycenter of its neighbors, it can be connected to a vertex with strictly bigger h -value. In turn, that vertex has to be active, and thus we can find a ***rising path*** from v to an exterior vertex, i.e., a path where the h values strictly increase. Similarly, one can find a ***falling path***.

Lemma 1.40. *For any inner vertex v , $\tau(v)$ is contained in the strict convex hull of the outer vertices.*

Proof. We first prove that $\tau(v)$ is contained in the convex hull of the outer vertices. Otherwise there exists a vertex v and an affine form h so that $h(\tau(v)) > 0$ but for all the outer vertices w , $h(\tau(w)) \leq 0$. Pick such a v of maximal h -value. Then all of its neighbors must have the same

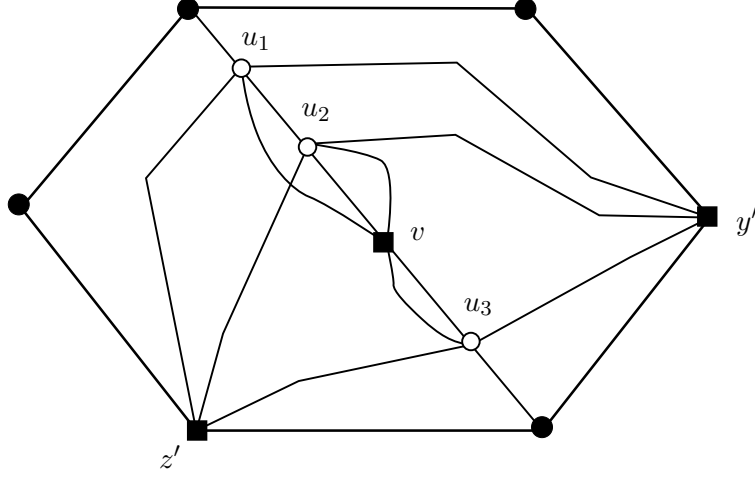


Figure 9: An inactive vertex leads to a $K_{3,3}$ subdivision.

value, i.e., v is inactive. We continue inductively: all their neighbors must have the same value, until by connectivity we reach an exterior vertex, and a contradiction.

So let us now assume that for some affine form h , we have $h(\tau(v)) = 0$, while h is negative for all the outer vertices except for two of them (since the starting embedding of the outer face is strictly convex). Denote by H the subgraph formed by inner vertices in $h^{-1}(0)$. H is non-empty since it contains v , and all the vertices in H are h -inactive. By 3-connectedness of G , H is connected to the rest of G in at least 3 ways, and is thus connected to at least one vertex such that h is negative. This means that at least one vertex in H is h -active, a contradiction. $\square$

Lemma 1.41. *For any affine form h , all the interior vertices are h -active.*

Proof. Assume otherwise that some interior vertex v is inactive. Pick a vertex w on the outer face so that $h(\tau(w)) > h(\tau(v))$. Then, by 3-connectedness, there exists three independent paths P_i between v and w . Denote by u_1, u_2 and u_3 the first active vertices on P_i . Note that the three of them satisfy $h(\tau(u_i)) = 0$. Then we take three rising paths from u_i to a vertex y of maximal h -value. While the three rising paths may intersect, it is easy to see (and asked as an exercise just below this proof) that from them, we can always find a vertex y' such that there are three independent paths from u_i to y' . Similarly, we take three falling paths from u_i to a vertex z of minimal h -value, and out of them we can find three independent paths to a vertex z' . Now we look at the paths between v, u_1, u_2 and u_3 , as well as the six other paths we have constructed, and recognize a subdivision of $K_{3,3}$, see Figure 9. This contradicts the planarity of the graph, and proves the lemma. $\square$

Exercise 1.42. Let u_1, u_2, u_3 and y be distinct vertices of a graph. Assume that for each i , there exists a path P_i connecting u_i to y which avoids all the other u_j for $j \neq i$. Then there exist three independent paths P'_i connecting u_i to a common vertex y' .

Now that every vertex has been proved to be active, we can prove that τ is an embedding locally:

Lemma 1.43. *Let uvx and uvy be two facial triangles incident to an edge uv of G not in C , then $\tau(x)$ and $\tau(y)$ are each on one side of any line through $\tau(u)$ and $\tau(v)$.*

Note that the lemma does not assume that $\tau(u) \neq \tau(v)$.

Proof. As usual, pick h an affine form whose kernel goes through $\tau(u)$ and $\tau(v)$. Pick two rising paths from $\tau(u)$ and $\tau(v)$ to an exterior vertex. (A subset of) These two rising paths and the edge uv , plus possibly some exterior edges, give rise to a cycle K for which all the h -values are positive. As in the proof of Lemma 1.40, all the vertices interior to K have positive h -value. In particular, one of x or y has positive h -value. Doing the same with falling paths, we get that one of x or y has negative h -value, which proves the lemma. $\square$

Say that a facial triangle is non-degenerate if its embedding has positive area. From this lemma, we conclude that all the facial triangles are non-degenerate. Indeed, this is clear for the triangles adjacent to outer edges, and this lemma shows that a triangle adjacent to a non-degenerate triangle is also non-degenerate. Therefore the claim follows by connectivity.

We can now conclude the proof of Theorem 1.37:

Proof. We have now proved that all the facial triangles are non-degenerate. Let us prove that the embeddings of two different triangles have distinct interiors. If not, let p be a point contained in two triangles t and t' . Throw a ray from p that avoids all the vertices $\tau(v)$. This ray exits t at some edge, where it enters a different triangle t_2 (by Lemma 1.43)). Continuing like that until we hit the outer face, we have a sequence $t, t_2, \dots, t_n$ where t_n is adjacent to the outer face. Likewise, the ray crosses t' and induces a sequence of triangles $t', t'_2, \dots, t'_m$ where t'_m is adjacent to the outer face. But the ray crosses the outer face at a given point, where there is a unique triangle incident to the outer edge. So $t_n = t'_m$, and then, following the ray back, we see that all of the triangles in the sequence are equal, including t and t' . $\square$

A nice corollary of this theorem is that we can always embed a planar 3-connected graph so that its faces are convex.

1.7 Minimum cut

Our last foray into algorithms for planar graphs concerns the computation of a minimum cut: given two vertices s and t , called **terminals**, on a planar graph $G = (V, E)$, we want to compute the minimum set of edges X so that removing X from E separates s and t .

Theorem 1.44. *Let $G = (V, E)$ be a cellularly embedded edge-weighted planar graph, and s and t be two distinct vertices of G . Then the problem of computing a minimum $s - t$ -cut of G can be solved in $O(n \log n)$ time.*

The basic idea of an efficient algorithm for min-cut on planar graphs is to look at it through the lens of duality: a cut on the primal graph separating s and t dualizes to a cycle in the dual graph separating the faces dual to s and t .

Proposition 1.45. *$X \subseteq E$ is an (s, t) -cut in G if and only if X^* contains the edge set of some cycle of G^* separating s and t .*

This proposition is considered obvious pretty much anywhere, but we will prove it, if only to emphasize that it does not hold on other surfaces (as usual, we will use the Jordan curve theorem).

Proof. The reverse direction is straightforward: if X^* contains a cycle of G^* separating s and t , then any path in G between s and t must cross this cycle, and thus X is an (s, t) -cut.

For the forward direction, we take X an (s, t) -cut in G , and choose C to be an inclusionwise minimal subset of X that is also an (s, t) -cut in G . We show that C^* is a cycle separating s

and t . By minimality of C , each vertex of G can be connected to either s or t without taking edges of C , and the two cases are exclusive. We label vertices with “S” or “T” depending on which one they are connected to. Moreover, for any edge in C , its two endpoints cannot have the same label, and for any other edge, its endpoints have the same label. So if we look at a face f of G adjacent to an edge of C , the labels on the facial walk on the face alternate only when the edge is in C , and thus there is an even number of edges of C adjacent to f . So C^* is an *Eulerian* subgraph of G^* , that is, a subgraph where each vertex has even degree. Pick any cycle in that subgraph. By the Jordan curve theorem, it separates s and t . By minimality, C^* is that cycle. $\square$

This proposition transforms a *combinatorial* problem into a *topological* one, namely, finding in the dual graph the shortest cycle enclosing a given face and not another given one. Even more topologically, if we *remove* the faces s^* and t^* from the dual graph, we obtain a surface homeomorphic to an annulus, and we look for the shortest cycle that goes around this annulus. However, this runs into an interesting technical issue: if s^* and t^* are adjacent, then removing s^* and t^* does not actually yield an annulus. Morally, this should not pose a problem: we just want to add an infinitesimally small buffer between s^* and t^* and we will be fine. One way to do this is to enlarge a tiny bit the set of curves that we look at. When working on the primal graph, we generally work with walks on the primal graph. When working on the dual graph, we work with walks on the dual graph, which correspond by duality to closed curves that are in *general position* with respect to G , i.e., they do not meet the vertices of G and cross the edges of G transversely. So our solution is to directly work in this setting of curves in general position with respect to G . The length of such a curve is defined to be the number of edges of G that it crosses. Note that this is a bit more general than just looking at walks on the dual graph: now we can define a pair of small curves in general position around s and t that are disjoint, even if s and t are adjacent in G . Yet from an algorithmic perspective, all the curves in general position can be pushed on the dual graph in a way that does not change the length, so any computation, for example shortest paths, can be made in the dual graph. In this new setting, Proposition 1.45 becomes:

Proposition 1.46. *Let γ be a simple closed curve in general position with respect to G , that separates s from t and that has minimal length among all such curves. Then the set of edges crossed by γ is a minimum (s, t) -cut in G .*

Recall that a simple closed curve on a sphere is an injective map $\gamma : \mathbb{S}^1 \rightarrow \mathbb{S}^2$.

Proof. Any path connecting s to t in G crosses γ , thus the set of edges crossed by γ is an (s, t) -cut. Conversely, a minimum (s, t) -cut dualizes to a cycle in G^* separating s and t , which corresponds to a simple closed curve γ in general position with respect to G , that separates s from t . $\square$

Now, we remove a small disk around s and t , getting an annulus A , and a portion of the graph G embedded on this annulus as in Figure 10. How do we compute a shortest closed curve that goes around the annulus A ? We can fix one point on each boundary, and compute a shortest path between these two points, and call it p . Then we show that some shortest closed curve does not cross p more than once, and thus can be found by a shortest path computation.

Lemma 1.47. *Some shortest closed curve separating the two boundaries of A is simple and crosses p exactly once.*

Proof. This is illustrated in the left of Figure 11. Let γ be such a curve. If we cut A along p , we obtain a topological disk D , where the path p got cut into two paths p' and p'' on the

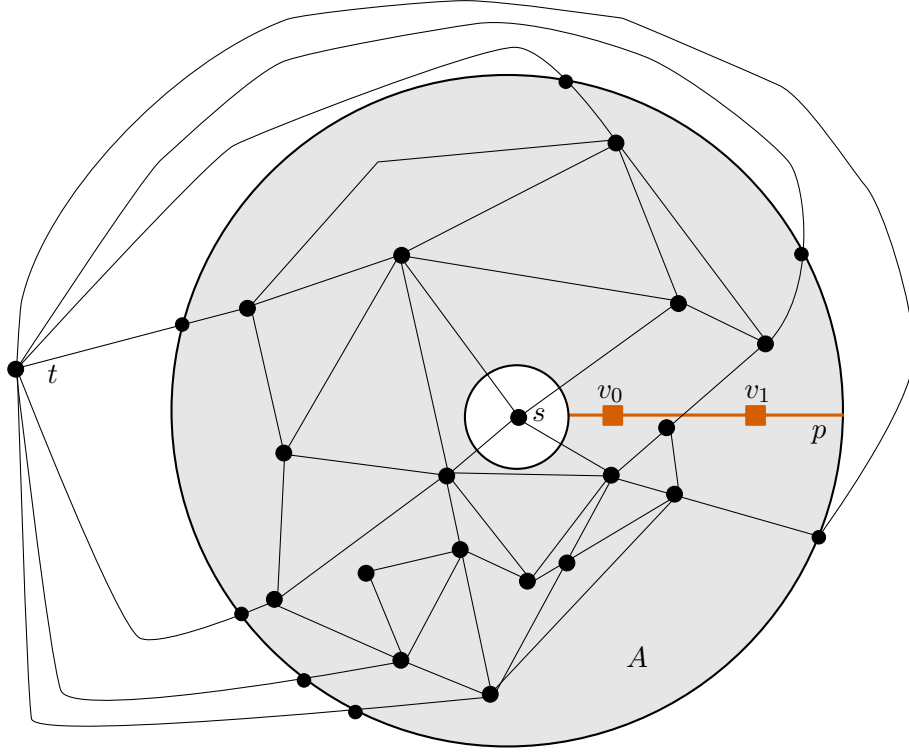


Figure 10: The annulus obtained after removing a small disk around s and t .

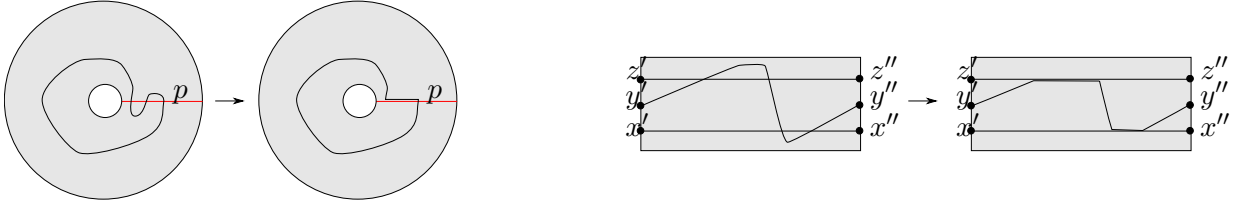


Figure 11: The two shortcutting arguments of Lemma 1.47

boundary. The curve γ , once cut on D , contains a simple path q connecting p' to p'' , otherwise γ would not separate s from t . Now we reglue D along p , and connect the two endpoints q_1 and q_2 of q by running parallel to p . We obtain a closed curve that is simple, crosses p exactly once and is no longer than γ since γ was connecting q_1 to q_2 as well. $\square$

From this we get a naive algorithm. We compute the shortest path p , which has some length k . We pick points $v_0, \dots, v_k$ on this path such that the subpath $[v_i, v_{i+1}]$ has length one. Then, cutting along p , each vertex v_i gets duplicated into v'_i and v''_i , and we compute all shortest paths between each v'_i and v''_i . Following all the lemmas, one of them is the dual of a minimal cut.

2 Surfaces

We now turn our attention to other topological spaces of dimension 2, and the graphs embedded thereon.

2.1 Definition and classification

A **surface** is a topological space locally homeomorphic to the plane, i.e., every point has an open neighborhood homeomorphic to $\mathbb{R}^2$. In this course, we restrict our attention to *compact* and *connected* surfaces. Behind this very wide-looking definition, there are actually only a few (yet infinitely many) different surfaces, and their classification is the topic of this subsection. Examples of surfaces are the sphere, the torus, and the projective plane, see Figure 12. As before, we consider spaces up to homeomorphism. It turns out that these are exhaustive in the sense that all the other surfaces can be built by gluing those together.

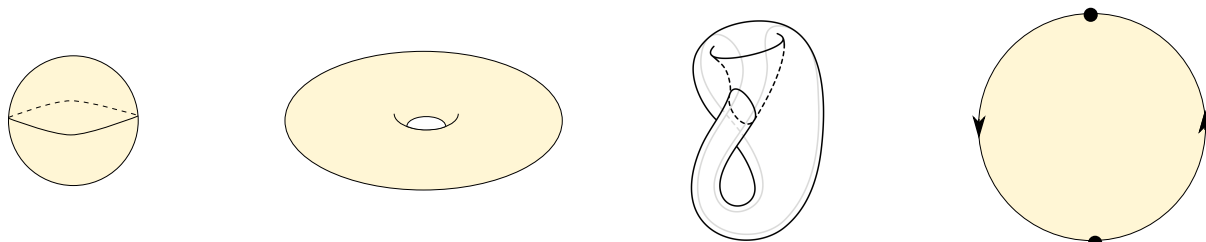


Figure 12: A sphere, a torus, a Klein bottle, and a projective plane (in the last one, the right boundary is identified to the left boundary in the direction indicated by the arrow).

In the first chapter, we saw how a connected graph embedded on the plane naturally cuts the plane into disks. Similarly, we will investigate and classify surfaces by the graphs embedded on them. The definition is the same as in the planar case: a graph $G = (V, E)$ is **embeddable** on a surface S if there exists an injective map $f : G \rightarrow S$, i.e., G can be drawn without crossings on S . A graph is **cellularly embedded** if every connected component of $S \setminus G$ is homeomorphic to an open disk.

Theorem 2.1 (Kerékjártó-Radó). *On any compact connected surface, there exists a cellularly embedded graph.*

Start of a proof. By definition of a surface, every point has an open neighborhood homeomorphic to the plane, or an open two-dimensional disk. By compactness, we can extract a finite covering out of this collection of open sets. Their boundaries form a family of simple closed curves embedded on the surface. If they cross finitely many times, the interiors of their intersections can be seen to be bounded by simple closed curves, and thus, by the Jordan-Schoenflies theorem (which applies in the small neighborhoods since they are homeomorphic to the plane), they bound disks. Hence we have cut the surface into disks, and we win. If they cross infinitely many times, we can tinker with them to reduce to the case of finite crossings. $\square$

This is more subtle than it appears: this sketch can be made correct in two and three dimensions, but not in higher dimensions, as there are example of higher dimensional manifolds that can not be *triangulated* (a higher dimensional analogue of our cutting into disks).

While we are not proving theorems, let us add the following one. A **triangulation** is a cellularly embedded graph where every face has degree 3. A refinement of a triangulation is obtained by either subdividing a face (adding a vertex in that face and edges adjacent to each vertex incident to that face), or an edge (adding a vertex on an edge and edges adjacent to each non-adjacent vertex in the two incident faces).

Theorem 2.2. *Any two triangulations of a surface have a common refinement.*

The same not-a-proof (does not) work(s): overlay the two triangulations, if they have a

finite number of intersections, we are done. Otherwise, we tinker things. Once again, this is quickly not true in higher dimensions.

For the reader disappointed in these two omissions (and I understand them), one alternate way to view this is that we are only looking at surfaces that are *defined* as disks glued together in a finite way, with two surfaces being isomorphic if they have a common refinement (this can be taken as a definition). Then the two omitted proofs show that we obtain the same surfaces as with the more topological definition, but if we are content within the purely combinatorial world, we do not need this equivalence.

We describe a graph cellularly embedded on a surface via *polygonal schemata*, which encodes the way that the disks are glued together. Starting from a cellularly embedded graph, we first name the edges and orient them in an arbitrary way. Each facial walk induces a word (considered up to cyclic permutation), where an edge e taken in the reverse direction is denoted by $\bar{e}$ or e^{-1} . Each such facial walk is called a *relation*. The polygonal scheme is the data of all these facial walks. Reciprocally, if one is given a collection of words where each letter of the alphabet appears exactly twice, one can interpret those as disks glued to each other, which together form a surface.

With Theorem 2.1 in hand, we are now ready to classify surfaces:

Theorem 2.3. *Every compact connected surface is homeomorphic to a surface given by one of the following polygonal schemata, each made of a single relation:*

1. $a\bar{a}$ (the sphere),
2. $a_1b_1\bar{a}_1\bar{b}_1 \dots a_gb_g\bar{a}_g\bar{b}_g$ for some $g \geq 1$,
3. $a_1a_1 \dots a_ga_g$ for some $g \geq 1$.

The surfaces of the first and second category are called *orientable*, those of the third category are *non-orientable*. The integer g is the *genus* of the surfaces. The second case corresponds to g tori glued together (this is called a *connected sum*), or equivalently, a sphere to which we have glued g handles. The third case corresponds to g projective planes glued together, or equivalently, to a sphere with g disks removed on which we have glued g *Möbius bands*. See Figure 13, and Figure 14 for the polygonal schemata depicted on the surfaces. Many non-trivial homeomorphisms are hidden behind this apparently simple classification, and it almost equally simple proof: for example, the theorem implies that the connected sum of two projective planes is homeomorphic to a Klein bottle, and that the connected sum of a torus and a projective plane is homeomorphic to the connected sum of three projective planes.

Proof. Let S be a compact connected surface, and let G be a graph embedded on S , which exists by Theorem 2.1. We iteratively remove each edge adjacent to two different face, until there is just a single face. For each edge adjacent to two different vertices, we contract it, and keep the multiple edges or loops that might result, until there is just a single vertex. We now have a graph with a single vertex and a single face embedded on S . If there are no more edges, by uncontracting once we obtain a sphere as in case 1. of the theorem. Thus there is at least one edge.

The single face induces a polygonal scheme with a single relation. The rest of the proof aims at transforming this single relation into one of the two forms of Theorem 2.3 via *cut-and-paste* operations:

- If the polygonal scheme has the form $aPaQ$ where P and Q are possibly empty words, then we can transform it into $bbP\bar{Q}$ by adding a new edge and removing a , see Figure 15.

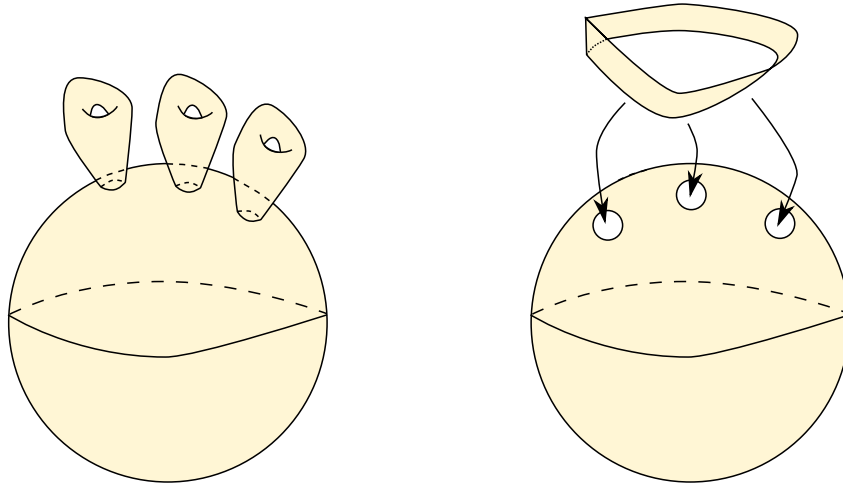


Figure 13: Attaching handles or Möbius bands to a sphere.

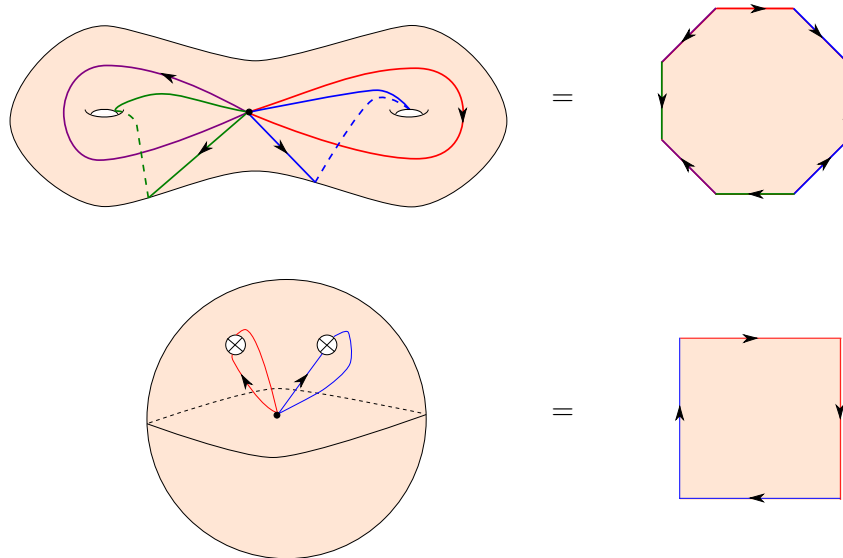


Figure 14: Polygonal schemata of the orientable and non-orientable surfaces. In the non-orientable cases, X denotes a disk on which a Möbius band has been glued.

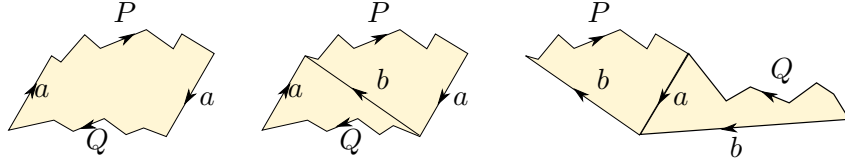


Figure 15: From $aPaQ$ to $bbP\bar{Q}$.

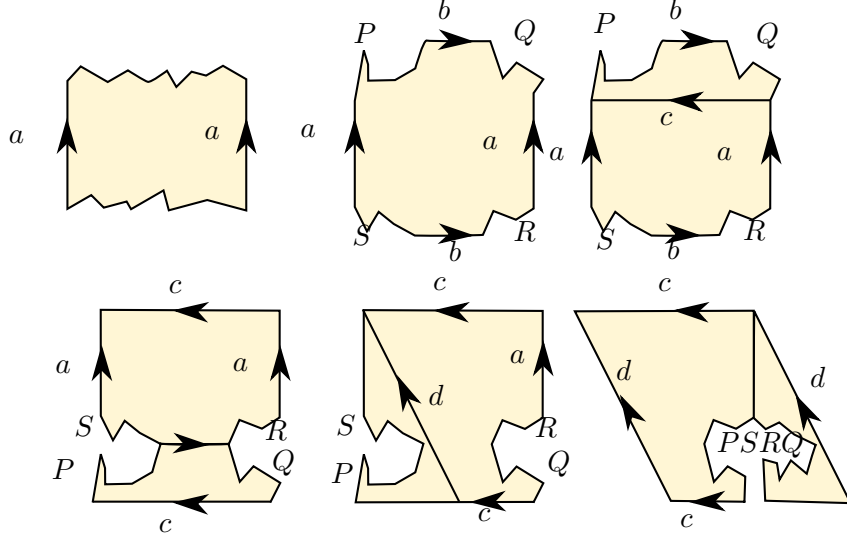


Figure 16: From $aPbQ\bar{a}R\bar{b}S$ to $cd\bar{c}dPSRQ$.

Inductively, we conclude that each pair of symbols with the same orientation appears consecutively in the polygonal scheme.

- If the polygonal scheme has the form $aU\bar{a}V$, then U and V must share an edge b since otherwise G' would have more than one vertex. By the preceding step, b must appear in opposite orientations in U and V , so we have the form $aU\bar{a}V = aPbQ\bar{a}R\bar{b}S$. This can be transformed into $cd\bar{c}dPSRQ$, as pictured in Figure 16. Inductively, at the end of this step the relation is a concatenation of blocks of the form aa or $ab\bar{a}\bar{b}$. If all the blocks are of one of these types, we are in case 2 or 3 and we are done.
- Otherwise, the relation has a subword of the form $aabc\bar{b}\bar{c}$. This can be transformed into $\bar{d}\bar{c}\bar{b}\bar{d}\bar{b}\bar{c}$, as in Figure 17, and then using the first step again this can be transformed into $eeffgg$. Inductively, we obtain a relation of the form 3.

This concludes the proof. □

The *Euler characteristic* of a surface S is defined to be $v - e + f$, where v , e and f are the numbers of vertices, edges and faces of a cellularly embedded graph on S . The fact that this actually does not depend on the graph is the object of the following proposition:

Proposition 2.4. *For any graph G cellularly embedded on S with v vertices, e edges and f faces, the value $v - e + f$ is the same.*

It is tempting to prove this using Theorem 2.3: any cellularly embedded graph can be transformed to one of the three graphs stipulated by the theorem, in a way that does not change the Euler characteristic. But to conclude, we need to prove that two different outputs of the theorem are not homeomorphic, which we have not done yet.

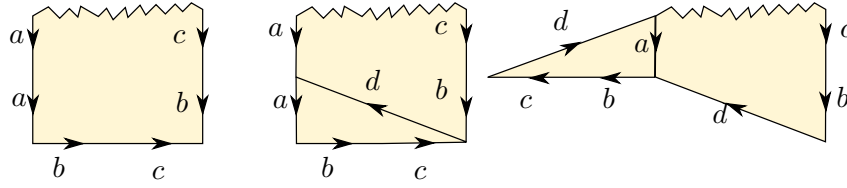


Figure 17: From $abc\bar{c}\bar{b}\bar{c}$ to $\bar{d}\bar{c}\bar{b}\bar{d}\bar{b}\bar{c}$.

Proof. We pick two graphs cellularly embedded on S . We add edges until they are both triangulations, which can easily be seen not to change the Euler characteristic. By Theorem 2.2, they have a common refinement. The proof follows from the fact that subdividing edges or faces does not change the Euler characteristic. $\square$

A look at the graphs corresponding to the cases 1, 2 and 3 of Theorem 2.3 shows that the sphere has Euler characteristic 2 (which should not come as a surprise after the first chapter), the orientable surface of genus g has Euler characteristic $2 - 2g$, and the non-orientable surface of genus g has Euler characteristic $2 - g$.

Proposition 2.5. *A surface is orientable if and only if for any cellularly embedded graph G , the boundaries of the faces can be oriented so that each edge appears in opposite directions in the two adjacent faces.*

Proof. First note that the property of the proposition is invariant under cut and pasting, as well as edge or face subdivision. Since the polygonal scheme of orientable surfaces satisfies the proposition, this proves the first implication. For the reverse implication, it suffices to observe that for non-orientable surfaces, such orientations of the faces are not possible. $\square$

As a corollary, a non-orientable surface is never homeomorphic to an orientable surface. Since all the orientable surfaces (respectively non-orientable surfaces) have a different Euler characteristic and the Euler characteristic is a topological invariant, this shows that all the surfaces given by Theorem 2.3 are pairwise non-homeomorphic. Therefore we have found them all, and they are all different.

Exercise 2.6. 1. Provide an example of two cellular embeddings of the same graph G on two different surfaces.

2. What is the surface described by the polygonal schemes $abc\bar{a}\bar{b}\bar{c}$? $abcd\bar{a}\bar{b}\bar{c}\bar{d}$?

We conclude with a few remarks:

Surfaces with boundary: It is often very convenient to also consider surfaces where points can be homeomorphic either to $\mathbb{R}^2$ or to an open half-space $\mathbb{R}^2 \cap \{x \geq 0\}$. These are surfaces with boundary, where the **boundary** is the set of points homeomorphic to an open half-space. Examples are the disk, the annulus, the torus with a disk removed, etc. A similar classification theorem holds for those: surfaces are still classified by their genus, their orientability, but also by the number of connected components of their boundary. It can be proved by gluing disks on the boundary to reduce to the usual case of surfaces. For G a graph embedded on a surface S (in particular for G a simple closed curve), one can **cut** S along G and obtain a surface with boundary which we denote by $S \bowtie G$.

Maps: As in the planar case, the actual description of a cellularly embedded graph can be made purely combinatorial. This is already the case with the polygonal schemes, and can be made more convenient algorithmically by using flags, in exactly the same way as in the planar case.

Duality: A graph cellularly embedded on a surface naturally induces a dual graph, as in the planar case.

Subdivisions and minors: There is no analogue of Kuratowski's theorem for surfaces, but there is an analogue for Wagner's theorem. However, the proofs are non-constructive, and the precise list of minors is unknown except for the projective plane.

2.2 The space of cycles and cuts

It is of interest to count cycles and cuts. Let G be a graph embedded on a surface S , with its set of vertices, edges and faces. We think of those as being 0-dimensional, 1-dimensional and 2-dimensional objects. A **k -chain**, for $k = 0, 1$ or 2 is a subset respectively the set of vertices, edges or faces. We will denote these space of subspace by C_0, C_1 or C_2 these space of k chain. On these space there is a vectorial space structure where the sum of $A \in C_n$ and $B \in C_n$ and scalar product are given by

$$A \oplus B = (A \cup B) \setminus (A \cap B) \quad 2A = A \oplus A = \emptyset \quad A \oplus \emptyset = A \quad 3A = A.$$

A basis of C_0 is given by the set of singletons listing all the vertices $C_0 = Vect(\{p_0\}, \dots, \{p_n\})$ and $\dim C_0 = |V|$, $\dim C_1 = |E|$, $\dim C_2 = |F|$. We think the k -chain as being an element in a vector space over $\mathbb{Z}_2$, the set of integers mod 2. Therefore, chains can be added, using the rule $1 + 1 = 0$. We define the boundary of an edge to be the sum of its endpoints, and the boundary of a face to be the sum of its boundary edges. These two boundary maps extend by linearity on the whole spaces C_1 and C_2 , defining linear maps

$$\partial_1 : C_1 \rightarrow C_0 \text{ and } \partial_2 : C_2 \rightarrow C_1.$$

The matrix associate to ∂_1 has $|V|$ lines and $|E|$ rows. A 1-chain is a **cycle** if its boundary is trivial, and it is a **boundary** if it is the boundary of some 2-chain. Cycles are generally denoted by $Z_1 = \ker \partial_1$, and boundaries by $B_1 = \text{Im} \partial_1$. Convince yourself that the boundary of a boundary is empty i.e. $\partial_2 \circ \partial_1 = 0$.

We define **cuts** in a graph G by the set of edges separating a subset of vertices from the rest of the graph. For $U \subset C_0$. let

$$\text{Cut}(U) = E_G \cap (U \times \bar{U}) = \{e \in E_G; |e \cap U| = 1\}$$

Then $\text{Cut} : C_0 \rightarrow C_1$ is a linear map with $\text{Cut}(A \oplus B) = \text{Cut}A \oplus \text{Cut}B$. The matrix associate tho th elinear map Cut is given by $\partial_1^T : C_0 \rightarrow C_1$ and $\partial_1^T(U)$ count edges incident to U . In particular if U is a connected component of G , then $\partial_1^T(U) = \emptyset = 0$ and $\dim \ker \partial_1^T = c$.

We define the **space of cuts by** $C_G^* = \text{Im } \partial_1^T$ and the **space of cycle** $C_G = \ker \partial_1 = \{X \in C_1; \partial_1 X = 0\}$. If the edge $e = p_i p_j$, then $\partial_1 e = \{p_i\} \oplus \{p_j\} \in C_0$. If X is the set of edges of a cycle then each vertex is counted twice:

$$\partial_1 X = \{p_j\} \oplus \{p_{j+1}\} \oplus \dots \oplus \{p_j\} = \emptyset = 0.$$

Since for $X \in C_G$ is a collection of vertices with even degree, we have that C_G is generated by Euler's circuit. A basis of C_G is given by a linearly independent set of cycles. If we consider a spanning tree T and if we add an edge we obtain a cycle in $T \cup \{e\}$. Hence the set $\{e_1, e_2, \dots, e_k\} = E \setminus T$ is the complementary set of edges of T in G , we have an independant set of cycle corresponding to the cycle contains into $T \cup \{e_i\}$. If c is the number of connected component of G , then $n - c$ is the number of edges of the spanning tree T and $\dim C_G = m - (n - c)$. Since

$$\dim C_G^* = \dim \text{Im} \partial_1^T = \dim C_0 - \dim \ker \partial_1^T = n - c$$

we have

$$\dim C_G + \dim C_G^* = \dim C_1.$$

For the planar graph we have

Theorem 2.7. *Let G be a 2-connected planar graph embedded in the plane. The bounded faces cycles of G form a basis for cycles in G .*

To prove this theorem it is enough for a cycle to use Jordan curve theorem which define an outside and inside disk. This disk is a concatenation of faces.

Theorem 2.8. *IF G is planar and 3-connected then C_G is isomorphic to $C_{G^*}^*$.*

Proof. It is enough to remark that

$$\dim C_G = |E| - (n - 1) = |F| - 1 = \dim C_{G^*}^*$$

□

2.3 Betti numbers of a planar graph

Definition 2.9. *A complex chain is a sequence of linear transformation $\partial_i : C_i \rightarrow C_{i-1}$ such that $\partial_{i-1} \circ \partial_i = 0$ where ∂_i is the boundary map and*

$$\partial_i \sigma_i = \sum \sigma_{i-1} \text{ where } \sum \sigma_{i-1} = \text{cycle forming the boundary of } \sigma_i$$

The i -th $\mathbb{Z}_2$ -homology group H_i is the vector space

$$H_i(C, \mathbb{Z}_2) := \mathbb{R}^{\text{Null } \partial_i - \text{Rank } \partial_{i+1}}$$

where $\text{Null } \partial_i - \text{Rank } \partial_{i+1} = \dim \ker \partial_i - \dim \text{Im } \partial_{i+1}$. The i -th betti number of the graph G is define by $b_i = \text{Null } \partial_i - \text{Rank } \partial_{i+1}$

Theorem 2.10. *Let G a graph and $C_0 \cup C_1 \cup C_2$ a complex chain. Then*

$$|V| - |E| + |F| = b_0 - b_1 + b_2$$

Proof. We have $|V| = \text{Null } \partial_0 + \text{Rank } \partial_0$, $|E| = \text{Null } \partial_1 + \text{Rank } \partial_1$ and $|F| = \text{Null } \partial_2 + \text{Rank } \partial_2$. Now with $\text{Rank } \partial_0 = \text{Rank } \partial_3 = 0$:

$$|V| - |E| + |F| = \text{Null } \partial_0 + \text{Rank } \partial_0 - \text{Null } \partial_1 - \text{Rank } \partial_1 + \text{Null } \partial_2 + \text{Rank } \partial_2$$

$$|V| - |E| + |F| = (\text{Null } \partial_0 - \text{Rank } \partial_1) - (\text{Null } \partial_1 - \text{Rank } \partial_2) + (\text{Null } \partial_2 - \text{Rank } \partial_3) = b_0 - b_1 + b_2$$

□

2.4 Some topological algorithms

The main difference between the plane and more complicated surfaces is that there is no Jordan curve theorem outside of the plane, and thus that some simple closed curves can be non-separating. Even among separating curves, there are differences, as is outlined in the following exercise:

Exercise 2.11. • Show that on an orientable surface of genus $g > 0$, there are exactly $\lfloor g/2 \rfloor + 2$ non-homeomorphic⁵ simple closed curves.

- (Harder) How many homeomorphism classes of curves are there on a non-orientable surface?

A standard criterion to differentiate closed curves on surfaces is *homotopy*. Two closed curves γ_1 and γ_2 are *homotopic* if there is a homotopy between them⁶: a continuous map $h : [0, 1] \times \mathbb{S}^1 \rightarrow S$ so that $h(0, \cdot) = \gamma_1$ and $h(1, \cdot) = \gamma_2$. The analogues of the nice curves in the planar case are the contractible curves: A simple closed curve γ on a surface S is *contractible* if it is homotopic to a trivial map at a point p . The following lemma, which could be taken as a definition, explains why:

Lemma 2.12. *A simple closed curve is contractible if and only if it bounds a disk.*

Half-proof: The reverse direction is immediate. For the forward direction, the subtlety comes from the fact the homotopy could introduce self-crossings. It turns out to be the case that such self-crossings are actually not needed, but we will not prove this. $\square$

Conversely, non-contractible curves are the main object of interest on surfaces. Note that non-contractible curves can be separating. More generally,

We will address two basic problems related to non-trivial curves:

1. How to test whether a closed curve is contractible?
2. How to compute the shortest non-contractible, or non-separating curve?

These algorithmic questions will also serve as an opening for the end of the course, as we use them to illustrate how classical tools from algebraic topology (universal covers, homology) impact the design of algorithms for topological problems on surface-embedded graphs.

2.5 Homotopy testing

To test whether a closed curve on a surface S is contractible, if the curve is simple, we can simply cut along it and see whether one of the components is a disk. This works because of Lemma 2.12. But if the curve is not simple, the problem is significantly less obvious to tackle. This is a good excuse to do some more topology. We will focus on getting polynomial algorithms. With quite some work, everything can actually be made linear.

The main strategy that we use is to use an auxiliary space from S , called the *universal cover* (or at least a portion of it), which differentiates in a natural way curves that are not homotopic. A *covering space* $(\hat{S}, p)$ for a surface S is a topological space with a continuous map $p : \hat{S} \rightarrow S$ that is a local homeomorphism: any point x in S has an open neighborhood U so that $p^{-1}(U)$ is a disjoint union of open sets U_i which are all homeomorphic to U . The *universal cover* is a covering space where *every* simple closed curve is contractible – such a space is called *simply connected*. Such a universal cover is unique in some (natural) sense, though we will not prove this nor use it.

This abstract definition makes much more sense when looking at specific examples, see Figure 18 for some illustrations:

⁵Technical remark: any two simple closed curves are homeomorphic (since they are all homeomorphic to $\mathbb{S}^1$). Here, we mean “homeomorphism of the surface that sends one curve to the other one”.

⁶This definition requires curves to be oriented, we will often disregard this by considering two curves to be homotopic if some orientation makes them homotopic.

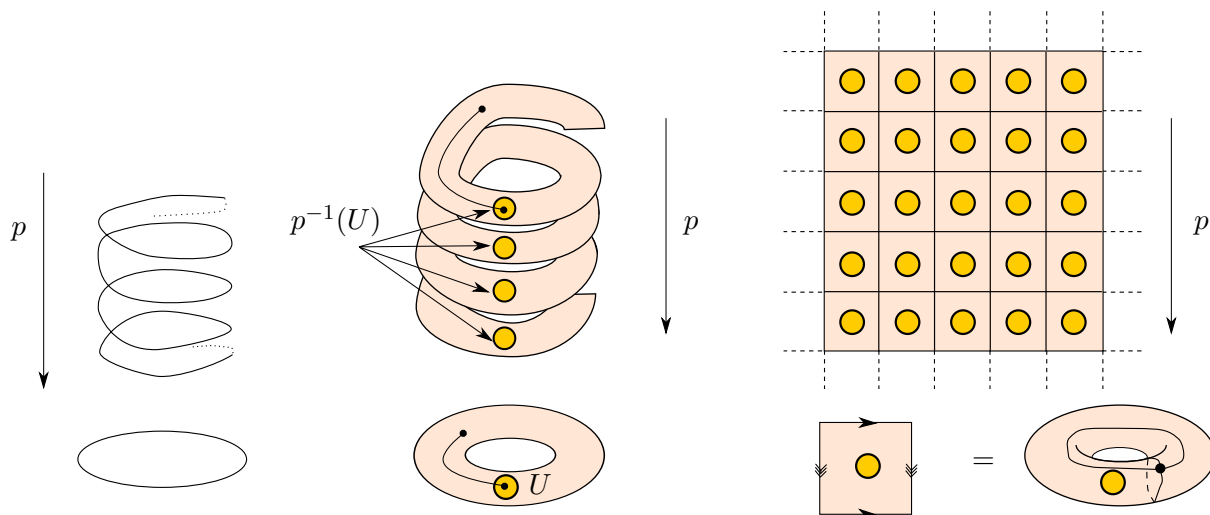


Figure 18: The universal covers of the circle, the annulus and the torus.

- The universal cover of a circle $\mathbb{S}^1$ is an infinite spiral over this circle.
- The universal cover of an annulus $\mathbb{A}$ is an infinite strip over this annulus.
- The universal cover of a sphere is the sphere itself.
- The universal cover of a torus is the plane $\mathbb{R}^2$ with a tiling into squares.

Exercise 2.13. What is the universal cover of the projective plane?

Since the map p of a universal cover is a local homeomorphism, every path, or closed curve γ on S , once a specific preimage of one of its points has been chosen, can be traced on $\hat{S}$, yielding a *lift* $\hat{\gamma}$. This is pictured in Figure 18 in the annulus example. But this lift might not be a closed curve. The following lemma shows that it is a closed curve exactly when it is contractible:

Lemma 2.14. *A closed curve γ on a surface lifts in the universal cover to a closed curve if and only if it is contractible.*

Proof. The universal cover is simply connected, and thus each closed curve there can be contracted to a point. This contraction projects via p to a contraction on S , and thus a closed curve on S that lifts to a closed curve must be contractible. Conversely, the lift of a trivial curve is clearly a closed curve, and the contraction on S lifts to a contraction in $\hat{S}$. $\square$

Surfaces with boundary: For a surface with boundary, one can compute a portion of the universal cover in the following way. Starting from a graph G embedded on a surface with boundary, we first contract and remove edges until there is a single face and all the vertices are on the boundary. Denote by F the remaining edges. We cut along them, thus getting a disk D . The edges of F will act as *fences*: whenever we cross such an edge, and nothing has been put yet on the other side, we glue a new copy of D . This process is naturally infinite, but is already useful for testing contractibility of a curve γ . First, observe that when an edge of G gets removed, one can reroute γ without changing its homotopy type, which might increase its length by an $O(g)$ factor. Then one can build the portion of the universal cover on which the lift of γ will live, trace the lift of γ on this portion and check whether the end point is the same as the starting point, see Figure 19. Thinking a bit more about this algorithm, it really only boils

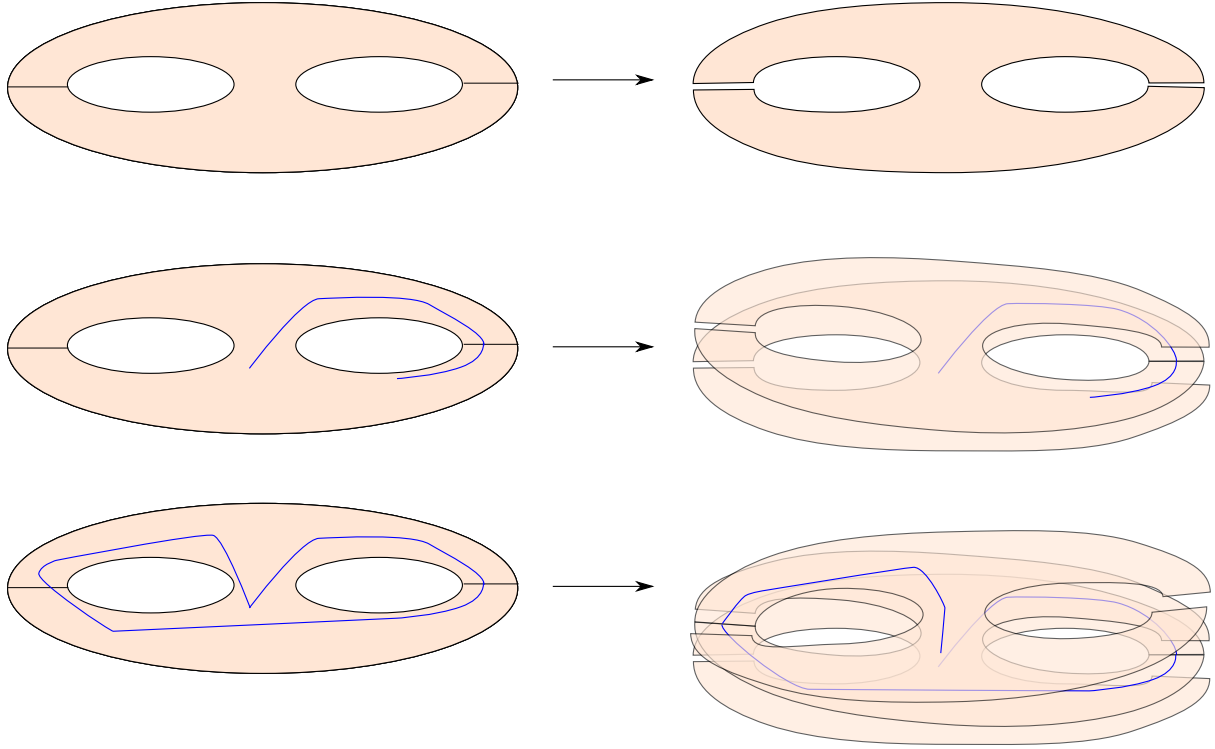


Figure 19: Lifting a closed curve on a disk with two holes. Note that in the cover, the path starts on the first “floor” and ends on the third floor. Hence the curve is not contractible

down to words: it amounts to writing the word formed by the edges of F that we intersect, with a $\cdot^{-1}$ if the orientation is reversed, and checking whether the word we obtain can be reduced to a trivial word by reducing subwords of the kind aa^{-1} . We clearly get a polynomial algorithm.

Orientable surfaces without boundary: We first look at the easy example of the torus. In this case, the universal cover is $\mathbb{R}^2$ tiled by squares, and the boundaries of the square can be taken to be any pair of edges (a, b) in a single vertex, single face graph embedded on the torus. One can test the contractibility of a curve by walking on this tiling of $\mathbb{R}^2$ and checking whether we come back to the same vertex at the end of the walk, see Figure 20. We emphasize here the main difference compared to the case with a boundary: after following the path for some time, we will sometimes come back to the same tile in $\mathbb{R}^2$ despite having never backtracked. Such a case happens in the example. Here again, in practice, this is a problem on words: this amounts to counting how many times the walk crosses a and b (counting the crossings positively or negatively depending on how we cross it) and checking whether at the end of the walk, those algebraic counts are zero.

For the more general case, as in the start of the classification of surfaces, up to removing and contracting edges, we can obtain from any graph a graph with a single vertex and a single face, which is often called a *system of loops*. By Euler’s formula, this system of loops has $4g$ loops. Cutting along it yields a $4g$ -gon. By analogy with the toroidal case, we want to look for the universal cover at a tiling of $\mathbb{R}^2$ where each tile is a $4g$ -gon, and every vertex has degree $4g$. This is not possible in a Euclidean way, but there are such *hyperbolic* tilings, which are tilings in a space of negative curvature, see Figure 21 for an example with the genus 2 surface. (One model of) Hyperbolic geometry is a geometry where the ambient space is an open disk (thus homeomorphic to $\mathbb{R}^2$) and the straight lines are arcs of circle which are orthogonal to the boundary of the disk. The reader can check on the picture that we indeed get a tiling of the

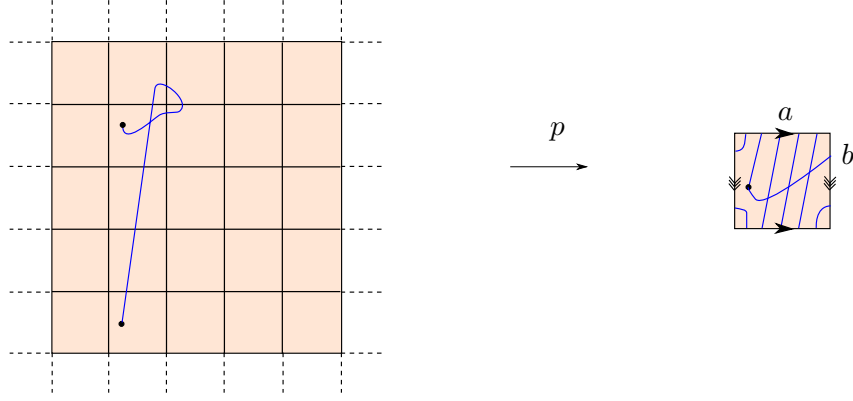


Figure 20: Lifting a closed curve on a torus. In the cover, it lifts to a path, hence the curve is not contractible

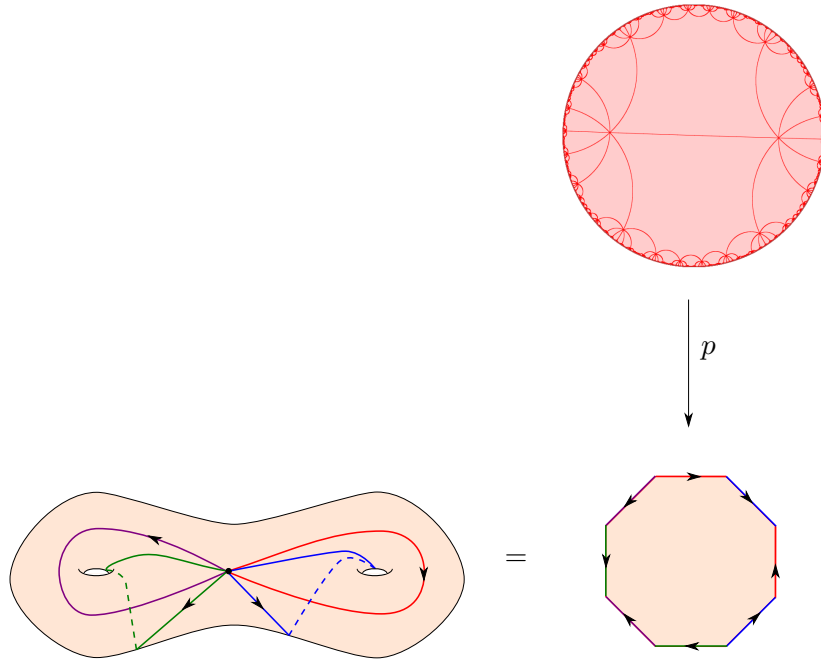


Figure 21: Covering a genus-two surface by a hyperbolic tiling of the open disk.

disk with octagons, eight of which meet at each vertex. Note that the definition of universal cover is purely topological, so the metric structure is just here at this stage to help us with intuition. One could try to trace on this hyperbolic tiling the walk and check whether we come back to a point, but now it is much less trivial to decide when we come back to the same tile. Instead, the following argument, dating back to Dehn, allows us to make progress locally until the curve is trivial:

A closed curve on a graph G which is a system of loops can be seen as a word on $A \cup A^{-1}$, where the alphabet A is the set of loops. A spur is a subword of the form aa^{-1} .

Lemma 2.15. *Let γ be a closed curve on a graph G which is a cellularly embedded system of loops on an orientable surface of genus at least 2. If γ has no spurs and is contractible, then it has a subpath consisting of more than half of a facial walk of G .*

Note that this is not true on a torus: indeed, this is a property that results from the hyperbolicity of the tiling.

One can prove this using Euler's formula and a bit of sweat, but since we have introduced a bit of geometry, it is a good occasion to explore a geometric version of the Euler formula argument. It relies on a discrete notion of curvature, which quantifies how non-Euclidean a space is, which we first introduce. Let D be a disk on which a graph is embedded. Around a vertex, between each pair of consecutive edges, there is a *corner*. For each corner c , let $\theta(c)$ be a positive number which we think of as the angle (in fractions of full turns) at this corner. In a Euclidean polygon, the sum of angles on a polygon with d sides is always $(d-2)\pi$, so with our normalizations, this becomes $d/2 - 1$. Therefore, it makes sense to define the curvature of a face $\kappa(f)$ to be

$$\kappa(f) = \sum_{c \in f} \theta(c) - \deg(f)/2 + 1.$$

Thus, a triangle where the sum of angles is less than π has negative curvature. Similarly, in usual space, the sum of angles around a vertex is 2π , or 1 when normalizing by full turns. Thus we define the curvature of an interior vertex $\kappa(v)$ is defined to be

$$\kappa(v) = 1 - \sum_{c \in v} \theta(c).$$

In these discrete terms, a vertex with negative curvature is a vertex with too much stuff around it. Finally, the curvature of a boundary vertex $\tau(v)$ is

$$\tau(v) = 1/2 - \sum_{c \in v} \theta(c).$$

A magical formula called the Gauss-Bonnet formula stipulates that on a surface, the sum of the curvature equals the Euler characteristic (up to some constant normalizing factor). In our setting, the discrete version is the following:

Lemma 2.16. *For a graph G embedded on a disk D , and any choice of angles $\chi(c)$ on its corners, if we denote by V_i and V_∂ its interior and boundary vertices, we have:*

$$\sum_{v \in V_i} \kappa(v) + \sum_{v \in V_\partial} \tau(v) + \sum_{f \in F} \kappa(f) = \chi(D) = 1.$$

Proof. Observe that in the three summands, the terms involving $\theta(c)$ cancel each other. What remains is $|V_i| + 1/2|V_\partial| + |F| - \sum_f \deg(f)/2$. The last sum counts the inner edges and half of the edges on the boundary, or equivalently, all the edges minus half the edges on the boundary. Since on the boundary of the disk, there are as many edges as vertices, this sums to $|V_i| + 1/2|V_\partial| + |F| - (|E| - |V_\partial|/2) = |V| - |E| + |F|$ which is exactly the Euler characteristic. $\square$

Of course, this generalizes to higher genus surfaces, with an identical proof. With this tool in hand, we proceed to the proof of Dehn's lemma.

Proof of Dehn's Lemma. If γ is contractible, it lifts to a closed curve in the universal cover. This closed curve might self-intersect, and thus partitions our tiling of $\mathbb{R}^2$ into disks (and the outer face). Therefore it suffices to prove that any disk D bounded by a closed curve on a $(4g, 4g)$ -tiling of $\mathbb{R}^2$ contains a subpath of length at least $2g + 1$ on the boundary of one tile. The idea of the proof is to choose the angles so that a lot of curvature has to happen on the boundary vertices, which will force the subpath that we are looking for.

Therefore, we set all the corners to have angle $1/4$. Then all the faces and all the interior vertices all have negative curvature. The vertices of the boundary have positive curvature if and only if they are adjacent to a single face, in which case they have curvature $1/4$. We call such a vertex *convex*. So by the combinatorial Gauss-Bonnet formula, there will be a lot of convex vertices:

$$\begin{aligned} \sum_{v \in V_i} \kappa(v) + \sum_{v \in V_\partial} \tau(v) + \sum_{f \in F} \kappa(f) &= 1 \\ |F|(1-g) + |V_i|(1-g) + |V_{\text{convex}}|/4 &\geq 1 \\ |V_{\text{convex}}| &\geq (4g-4)|F| + 4 \end{aligned}$$

So some face is adjacent to $4g-3$ convex vertices. Since edges adjacent to these convex vertices are on the boundary of the disk, $4g-3$ of these convex vertices must be consecutive. So some face has $4g-2$ consecutive edges on ∂D , which is strictly bigger than $2g$ for $g \geq 2$. $\square$

With Lemma 2.15 in hand, we can describe a combinatorial algorithm to test contractibility: after having reduced to a system of loops, we look at the word formed by the walk. We first inductively remove all the spurs. Then we scan it for a subword consisting of more than half of the facial walk of the system of loops. Whenever there is one, we replace it by the complementary part of the facial walk: this is a homotopy, and thus does not change the contractibility of the walk. Each of these changes reduces the complexity of the word, and we induct. By Lemma 2.15, if the closed curve is contractible, we will reach the trivial word. The complexity is clearly polynomial. With quite a lot of care, it can be made linear.

Zooming out a bit: In both variants, we have seen that the problem boils down to a problem on words: deciding whether a given word reduces to a trivial word under spur reduction (case with boundary), or with spur reductions and a more complicated relation. The underlying reason behind this is that the problem can be phrased in terms of combinatorial group theory. Indeed, the homotopy classes of loops on a surface, with the concatenation law, forms a group called the *fundamental group* of the surface, and a presentation for this group can be readily computed from a graph embedded on the surface: in the case with boundary, it is a free group with the fences as generators, and in the case without boundary, it is the single-relator group obtained with the system of loops as generators and the facial walk as a relation. Then the contractibility test amounts to testing the triviality of an element of this group. This perspective is easily misleading: in general testing triviality in a group defined by generators and relations is *undecidable*. The fact that this can be solved for the fundamental group of surface is therefore remarkable.

More generally, we can try to test whether two given curves are homotopic. This is more subtle, but can also be made in linear time, using similar tools.

3 Discrete calculus on graph and Steinitz theorem

3.1 Discrete calculus

The differential of a function $\phi : V \rightarrow \mathbb{R}^n$ is defined by the finite difference $d\phi(e) = \phi(p) - \phi(q)$ where the oriented edge $\vec{e} = qp$. To properly define the differential, we consider oriented space of edges $\vec{E}$ where $\vec{e} = p\vec{q} = -q\vec{p}$. The difference $d\phi : \vec{E} \rightarrow \mathbb{R}^n$ defines a one form with the property that $d\phi(-e) = -d\phi(e)$. The oriented boundary linear operator is $\partial : \vec{E} \rightarrow V$ where $\partial(q\vec{p}) = p - q$.

and $\partial^T : V \rightarrow \vec{E}$ the operator where $\partial^T p$ is the set of edges $\vec{e}$ with positive sign if $\vec{e}$ is pointing to p and negative sign if $\vec{e}$ is pointing out of p . This operator ∂^T induce a differential operator on the function ϕ , with

$$\partial^T \phi : \vec{E} \rightarrow \mathbb{R} \text{ with } \partial^T \phi(\vec{pq}) = \phi(q) - \phi(p).$$

Remark: If B is a matrix associate to ∂ , then B^T is the matrix associate to ∂^T .

Definition 3.1. A linear map $\omega : C_1 = E \rightarrow \mathbb{R}^n$ with $\alpha(-e) = -\alpha(e)$ define a one form on the graph G . The one form α is exact if there exists a function $\phi : V \rightarrow \mathbb{R}$ such that $d\phi = \alpha$. The one form α is closed if $\alpha(C) = 0$ for any cycle in G . In particular if C is a cycle associate to the set of edges $e_1, e_2, \dots, e_\ell$, we have

$$\sum_{i=1}^{\ell} \alpha(e_i) = 0$$

Hence if G is a 3-connected planar graph we have that a one form $\alpha : \vec{E} \rightarrow \mathbb{R}$ is close iff for any face $\phi \in F$, we have $\sum_{e_i \in \phi} \alpha(e_i) = 0$. (Any face is bounded by a cycle and any cycle bound a set of faces by the Jordan curve theorem). The equivalent Poincare Lemma in discrete calculus is

Theorem 3.2. If G is a cellularly embedded simply connected planar graph then any closed one form is exact. If $\phi : V \rightarrow \mathbb{R}$ satisfy $d\phi = \alpha$ then ϕ is defined up to a constant. If γ_1 is homotopic to γ_2 , then the value of ϕ does not depend on the path.

Proof. Assume that $\phi(p) = 0$. We construct the function ϕ by integrating α on each path. If γ is a path passing through the vertex $(p, p_1, \dots, p_n, q)$ lets, define

$$\phi(q) = \alpha(p_n q) + \alpha(p_{n-1} p_n) + \dots + \alpha(p_1 p)$$

If γ' is an other path passing through $(p, p'_1, \dots, p'_n, q)$, we need to prove that the value of ϕ does not depend in γ or γ' . If γ' is homotopic to γ , this mean that $\gamma \cup \gamma'$ bound a disk (this is always the case in simply connected domain) and the function ϕ is well define i.e.

$$\alpha(p_n q) + \alpha(p_{n-1} p_n) + \dots + \alpha(p_1 p) = \alpha(p'_n q) + \alpha(p'_{n-1} p_n) + \dots + \alpha(p'_1 p).$$

□

3.2 The Steinitz theorem

Every convex polytope in $\mathbb{R}^3$ has a set of *vertices* (extremal points) and *edges*. This is called the **1-skeleton** of the polytope. In this section, we will prove the following theorem.

Theorem 3.3 (Steinitz' theorem (1922)). Let $G = (V, E)$ be a graph without loops or multiple edges. Then G is a planar 3-connected graph if and only if it is the 1-skeleton of a convex polytope in $\mathbb{R}^3$.

The “if” part is the easiest direction: Imagine a big physical model of your convex polytope, where the faces and the interior of the polytope are transparent. If you sit outside the polytope, close enough to the center of a face, you will see no crossing between the edges. In other words, the 1-skeleton is planar. We omit the proof that the 1-skeleton of a 3-dimensional polytope is

3-connected; this can be proved directly without too much trouble, and follows from a more general theorem by Balinski (see the proof for example in Ziegler's book Section 3.5).

Let $\omega : E \rightarrow (0, \infty)$ be a function from the (undirected) edges of G to a set of strictly positive coefficients. (In the sequel, we could take ω to be constant, equal to one.)

Let $f : V \rightarrow \mathbb{R}^2$ be the corresponding Tutte equilibrium, where every vertex v is barycenter with coefficients $\omega_{vw_1}, \dots, \omega_{vw_m}$ of its neighbors $w_1, \dots, w_m$. We actually assume that $f : V \rightarrow \mathbb{R}^3$ maps the vertices into the plane $z = 1$ of $\mathbb{R}^3$.

To every *interior* face f of G we associate a vector q_f in $\mathbb{R}^3$. We choose an arbitrary interior face f_0 , for which $q_{f_0} = 0$. The other q_f 's are defined by the following formula: For every interior edge uv with left face f_1 and right face f_2 , we define

$$q_{f_1} = \omega_{uv}(f(u) \times f(v)) + q_{f_2} \quad (3.1)$$

where $\times$ denotes the cross-product in $\mathbb{R}^3$.

Lemma 3.4. *The vectors q_f are well-defined.*

Proof. First note that exchanging u exchanges f_1 and f_2 in (3.1), and thus gives $q_{f_2} = \omega_{vu}(f(v) \times f(u)) + q_{f_1}$, which rewrites $q_{f_2} = \omega_{uv}(-f(u) \times f(v)) + q_{f_1}$; this is exactly Equation (3.1).

Let v be a vertex of G , and $w_1, \dots, w_m$ be its neighbors. We get:

$$\sum_{i=1}^m \omega_{vw_i}(f(v) \times f(w_i)) = f(v) \times \left(\sum_{i=1}^m \omega_{vw_i}(f(w_i) - f(v)) \right) = 0.$$

Therefore, Equation (3.1) gives consistent vectors q_f for all faces around an interior vertex.

Now, starting from the initial face f_0 , we may define the value of q_f by choosing an arbitrary sequence of faces from f_0 . In other words, in the dual graph G^* , every path starting at f_0^* and ending at some face f gives a value of q_f . (Incidentally, this shows that the function q , if it exists, is unique.) We need to check that this value does not depend on the particular path chosen. For this purpose, consider two such paths p and p' in G^* . We may assume that p and p' use distinct vertex sets of G^* except at the endpoints f_0^* and f^* . Then by the Jordan curve theorem, p and p' enclose a set of faces of G^* . The result is proven by induction on the number of faces of G^* enclosed by p and p' : The case of one face is the previous paragraph. For the induction step, build one path p'' in G^* that is in-between p and p' , and apply induction. $\square$

We define a piecewise linear function g from the union of the interior faces to $\mathbb{R}$ by setting, for every point x in a face f , $g(x) = \langle x, q_f \rangle$.

Lemma 3.5. *This map g is well-defined.*

Proof. We only need to prove that, whenever x belongs to an edge uv incident with faces f_1 and f_2 , the value of $g(x)$ is the same, whichever face f_1 or f_2 we choose for the computation; in other words, $\langle x, q_{f_1} \rangle = \langle x, q_{f_2} \rangle$. By linearity it suffices to prove the result for $x = f(u)$ and $x = f(v)$.

$$\langle f(v), q_{f_1} \rangle = \langle f(v), \omega_{uv}(f(u) \times f(v)) + q_{f_2} \rangle = \langle f(v), q_{f_2} \rangle.$$

A similar computation holds for $f(u)$. $\square$

Lemma 3.6. *Let uv be an edge with left face f_1 and right face f_2 . Assume x is in the face f_1 . Then $\langle x, q_{f_1} \rangle < \langle x, q_{f_2} \rangle$.*

Proof.

$$\langle x, q_{f_2} \rangle - \langle x q_{f_1} \rangle = \omega_{uv} \langle x f(u) \times f(v) \rangle = \omega_{uv} \det(x, f(u), f(v)) > 0,$$

by our orientation convention (recall that the last coordinate of the points x , $f(u)$, and $f(v)$ is one) and the fact that $\omega_{uv} > 0$. $\square$

Sketch of proof of Theorem 3.3. Recall that the position of a vertex v is $f(v)$ in $\mathbb{R}^3$, actually in the plane $z = 1$. We just move vertically $f(v)$ to height $g(f(v))$. Let $F(v)$ be the new position. Let P be the convex hull of the $F(v)$. Lemma 3.6 implies that every interior edge uv is an edge of P , because every such edge is a “valley”; the same clearly holds for the exterior edges. It is clear that the lifts of all vertices on a given face are coplanar, and therefore each face of P is a convex polygon. Therefore, P is a convex polytope.

There is one subtlety, however: the vertices of the outer face are not necessarily coplanar; though, if the outer face is incident with three vertices, this condition is automatically satisfied. If G contains a triangle, we may have taken that triangle to be the outer face in the application of Tutte’s theorem. Thus, the only case that remains to be shown is when G contains no triangle.

We know that G^* , the dual graph of G , contains a triangle. Clearly, G^* is planar, and it can be shown that it is also 3-connected. We may therefore realize G^* as the 1-skeleton of a convex polytope. Now, a known construction, *polarity*, allows to transform a 3-polytope into another one, whose 1-skeleta are dual to each other Ziegler’s book Section. 2.3. So G is the 1-skeleton of a convex polytope as well. $\square$